CHAPTER 9

Electrostatics

9.1 Introduction ESBPH

In Grade 10, you learnt about the force between charges. In this chapter you will learn exactly how to determine this force and about a basic law of electrostatics.

Key Mathematics Concepts

- Ratio and proportion — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Scientific notation — Physical Sciences, Grade 10, Science skills

9.2 Coulomb's law ESBPJ

Like charges repel each other while unlike charges attract each other. If the charges are at rest then the force between them is known as the **electrostatic force**. The electrostatic force between charges increases when the magnitude of the charges increases or the distance between the charges decreases.

The electrostatic force was first studied in detail by Charles-Augustin de Coulomb around 1784. Through his observations he was able to show that the **magnitude** of the electrostatic force between two point-like charges is inversely proportional to the square of the distance between the charges. He also discovered that the **magnitude** of the force is proportional to the product of the charges. That is:

$$F \propto \frac{Q_1 Q_2}{r^2},$$

where Q_1 and Q_2 are the magnitudes of the two charges respectively and r is the distance between them. The magnitude of the electrostatic force between two point-like charges is given by *Coulomb's law*.

DEFINITION: *Coulomb's law*

Coulomb's law states that the magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

$$F = \frac{kQ_1 Q_2}{r^2},$$

The proportionality constant k is called the *electrostatic constant* and has the value: 9,0 $\times$ 10^9 N$\cdot$m$^2\cdot$C^{-2} in free space.

Similarity of Coulomb's law to Newton's universal law of gravitation.

Notice how similar in form Coulomb's law is to Newton's universal law of gravitation between two point-like particles:

$$F_G = \frac{Gm_1m_2}{d^2},$$

where m_1 and m_2 are the masses of the two point-like particles, d is the distance between them, and G is the gravitational constant. Both are inverse-square laws.

Both laws represent the force exerted by particles (point masses or point charges) on each other that interact by means of a field.

See video: 23YQ at www.everythingscience.co.za

Worked example 1: Coulomb's law

QUESTION

Two point-like charges carrying charges of $+3 \times 10^{-9}$ C and -5×10^{-9} C are 2 m apart. Determine the magnitude of the force between them and state whether it is attractive or repulsive.

SOLUTION

Step 1: Determine what is required

We are required to determine the force between two point charges given the charges and the distance between them.

Step 2: Determine how to approach the problem

We can use Coulomb's law to calculate the magnitude of the force. $F = k\frac{Q_1Q_2}{r^2}$

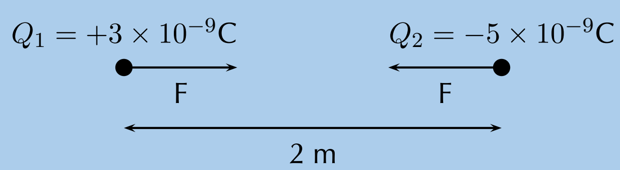
Step 3: Determine what is given

We are given:

- $Q_1 = +3 \times 10^{-9}$ C
- $Q_2 = -5 \times 10^{-9}$ C
- $r = 2$ m

We know that $k = 9{,}0 \times 10^9$ N·m^2·C^{-2}.

We can draw a diagram of the situation.



Step 4: Check units

All quantities are in SI units.

Step 5: Determine the magnitude of the force

Using Coulomb's law we have

$$\begin{aligned} F &= \frac{kQ_1Q_2}{r^2} \\ &= \frac{(9{,}0 \times 10^9)(3 \times 10^{-9})(5 \times 10^{-9})}{(2)^2} \\ &= 3{,}37 \times 10^{-8}\ \text{N} \end{aligned}$$

Thus the *magnitude* of the force is $3{,}37 \times 10^{-8}$ N. However since the point charges have opposite signs, the force will be attractive.

Step 6: Free body diagram

We can draw a free body diagram to show the forces. Each charge experiences a force with the same magnitude and the forces are attractive, so we have:

Next is another example that demonstrates the difference in magnitude between the gravitational force and the electrostatic force.

Worked example 2: Coulomb's law

QUESTION

Determine the magnitudes of the electrostatic force and gravitational force between two electrons 10^{-10} m apart (i.e. the forces felt inside an atom) and state whether the forces are attractive or repulsive.

SOLUTION

Step 1: Determine what is required

We are required to calculate the electrostatic and gravitational forces between two electrons, a given distance apart.

Step 2: Determine how to approach the problem

We can use:

$$F_e = \frac{kQ_1Q_2}{r^2}$$

to calculate the electrostatic force and

$$F_g = \frac{Gm_1m_2}{d^2}$$

to calculate the gravitational force.

Step 3: Determine what is given

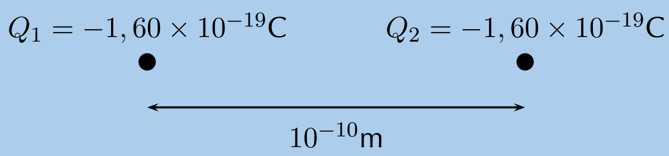
- $Q_1 = Q_2 = 1{,}6 \times 10^{-19}$ C (The charge on an electron)
- $m_1 = m_2 = 9{,}1 \times 10^{-31}$ kg (The mass of an electron)
- $r = d = 1 \times 10^{-10}$ m

We know that:

- $k = 9{,}0 \times 10^{9}$ N·m^2·C^{-2}
- $G = 6{,}67 \times 10^{-11}$ N·m^2·kg^{-2}

All quantities are in SI units.

We can draw a diagram of the situation.



Step 4: Calculate the electrostatic force

$$\begin{aligned} F_e &= \frac{kQ_1Q_2}{r^2} \\ &= \frac{(9{,}0 \times 10^{9})(1{,}60 \times 10^{-19})(1{,}60 \times 10^{-19})}{(10^{-10})^2} \\ &= 2{,}30 \times 10^{-8}\ \text{N} \end{aligned}$$

Hence the *magnitude* of the electrostatic force between the electrons is $2{,}30 \times 10^{-8}$ N. Since electrons carry like charges, the force is repulsive.

Step 5: Calculate the gravitational force

$$\begin{aligned} F_g &= \frac{Gm_1m_2}{d^2} \\ &= \frac{(6{,}67 \times 10^{-11})(9{,}11 \times 10^{-31})(9{,}11 \times 10^{-31})}{(10^{-10})^2} \\ &= 5{,}54 \times 10^{-51}\ \text{N} \end{aligned}$$

TIP
We can apply Newton's third law to charges because two charges exert forces of equal magnitude on one another in opposite directions.

TIP
Choosing a positive direction
When substituting into the Coulomb's law equation, one may choose a positive direction thus making it unnecessary to include the signs of the charges. Instead, select a positive direction. Those forces that tend to move the charge in this direction are added, while forces acting in the opposite direction are subtracted.

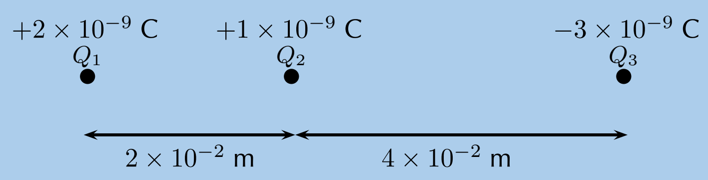
The magnitude of the gravitational force between the electrons is $5{,}54 \times 10^{-51}$ N. Remember that the gravitational force is always an attractive force.

Notice that the gravitational force between the electrons is **much smaller** than the electrostatic force.

Worked example 3: Coulomb's law

QUESTION

Three point charges are in a straight line. Their charges are $Q_1 = +2 \times 10^{-9}$ C, $Q_2 = +1 \times 10^{-9}$ C and $Q_3 = -3 \times 10^{-9}$ C. The distance between Q_1 and Q_2 is 2×10^{-2} m and the distance between Q_2 and Q_3 is 4×10^{-2} m. What is the net electrostatic force on Q_2 due to the other two charges?



SOLUTION

Step 1: Determine what is required

We need to calculate the net force on Q_2. This force is the sum of the two electrostatic forces - the forces between Q_1 on Q_2 and Q_3 on Q_2.

Step 2: Determine how to approach the problem

- We need to calculate the two electrostatic forces on Q_2, using Coulomb's law.
- We then need to add up the two forces using our rules for adding vector quantities, because force is a vector quantity.

Step 3: Determine what is given

We are given all the charges and all the distances.

Step 4: Calculate the magnitude of the forces.

Force on Q_2 due to Q_1:

$$\begin{aligned} F_1 &= k\frac{Q_1 Q_2}{r^2} \\ &= (9{,}0 \times 10^9)\frac{(2 \times 10^{-9})(1 \times 10^{-9})}{(2 \times 10^{-2})^2} \\ &= (9{,}0 \times 10^9)\frac{(2 \times 10^{-9})(1 \times 10^{-9})}{(4 \times 10^{-4})} \\ &= 4{,}5 \times 10^{-5}\text{ N} \end{aligned}$$

Force on Q_2 due to Q_3:

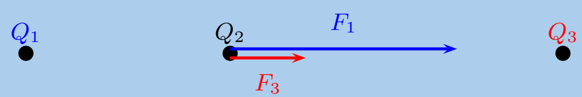
$$\begin{aligned} F_3 &= k\frac{Q_2 Q_3}{r^2} \\ &= (9{,}0 \times 10^9)\frac{(1 \times 10^{-9})(3 \times 10^{-9})}{(4 \times 10^{-2})^2} \\ &= (9{,}0 \times 10^9)\frac{(1 \times 10^{-9})(3 \times 10^{-9})}{(16 \times 10^{-4})} \\ &= 1{,}69 \times 10^{-5}\text{ N} \end{aligned}$$

Step 5: Vector addition of forces

We know the force magnitudes but we need to use the charges to determine whether the forces are repulsive or attractive. It is helpful to draw the force diagram to help determine the final direction of the net force on Q_2. We choose the positive direction to be to the right (the positive x-direction).

The force between Q_1 and Q_2 is repulsive (like charges). This means that it pushes Q_2 to the right, or in the positive direction.

The force between Q_2 and Q_3 is attractive (unlike charges) and pulls Q_2 to the right.



Therefore both forces are acting in the positive direction.

Therefore,

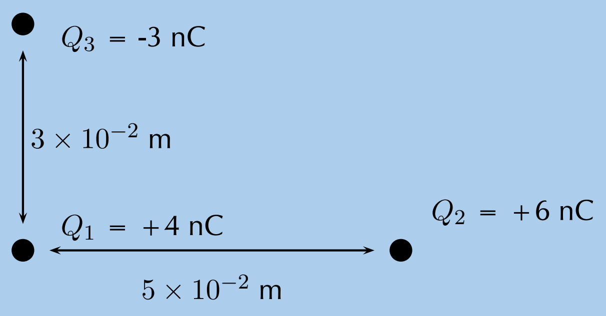
$$\begin{aligned} F_R &= 4{,}5 \times 10^{-5}\text{ N} + 1{,}69 \times 10^{-5}\text{ N} \\ &= 6{,}19 \times 10^{-5}\text{ N} \end{aligned}$$

The resultant force acting on Q_2 is $6{,}19 \times 10^{-5}$ N to the right.

Worked example 4: Coulomb's law

QUESTION

Three point charges form a right-angled triangle. Their charges are $Q_1 = 4 \times 10^{-9}$ C $=$ 4 nC, $Q_2 = 6 \times 10^{-9}$ C $=$ 6 nC and $Q_3 = -3 \times 10^{-9}$ C $= -3$ nC. The distance between Q_1 and Q_2 is 5×10^{-2} m and the distance between Q_1 and Q_3 is 3×10^{-2} m. What is the net electrostatic force on Q_1 due to the other two charges if they are arranged as shown?



SOLUTION

Step 1: Determine what is required

We need to calculate the net force on Q_1. This force is the sum of the two electrostatic forces - the forces of Q_2 on Q_1 and Q_3 on Q_1.

Step 2: Determine how to approach the problem

- We need to calculate, using Coulomb's law, the electrostatic force exerted on Q_1 by Q_2, and the electrostatic force exerted on Q_1 by Q_3.
- We then need to add up the two forces using our rules for adding vector quantities, because force is a vector quantity.

Step 3: Determine what is given

We are given all the charges and two of the distances.

Step 4: Calculate the magnitude of the forces.

The magnitude of the force exerted by Q_2 on Q_1, which we will call F_2, is:

$$\begin{aligned} F_2 &= k\frac{Q_1Q_2}{r^2} \\ &= (9{,}0 \times 10^9)\frac{(4 \times 10^{-9})(6 \times 10^{-9})}{(5 \times 10^{-2})^2} \\ &= (9{,}0 \times 10^9)\frac{(4 \times 10^{-9})(6 \times 10^{-9})}{(25 \times 10^{-4})} \\ &= 8{,}630 \times 10^{-5}\ \text{N} \end{aligned}$$

The magnitude of the force exerted by Q_3 on Q_1, which we will call F_3, is:

$$\begin{aligned} F_3 &= k\frac{Q_1Q_3}{r^2} \\ &= (9{,}0 \times 10^9)\frac{(4 \times 10^{-9})(3 \times 10^{-9})}{(3 \times 10^{-2})^2} \\ &= (9{,}0 \times 10^9)\frac{(4 \times 10^{-9})(3 \times 10^{-9})}{(9 \times 10^{-4})} \\ &= 1{,}199 \times 10^{-4}\ \text{N} \end{aligned}$$

Step 5: Vector addition of forces

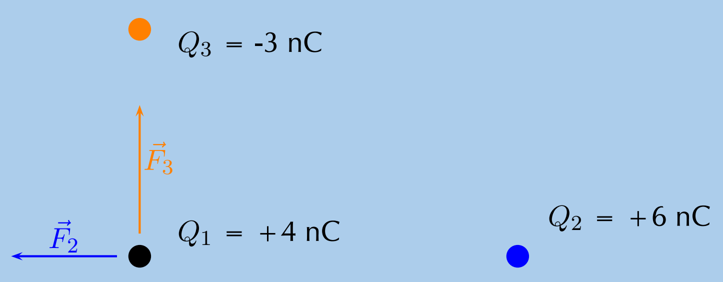
This is a two-dimensional problem involving vectors. We have already solved many two-dimensional force problems and will use precisely the **same procedure** as before. Determine the vectors on the Cartesian plane, break them into components in the x- and y-directions, and then sum components in each direction to get the components of the resultant.

We choose the positive directions to be to the right (the positive x-direction) and up (the positive y-direction). We know the magnitudes of the forces but we need to use the signs of the charges to determine whether the forces are repulsive or attractive. Then we can use a diagram to determine the directions.

The force between Q_1 and Q_2 is repulsive (like charges). This means that it pushes Q_1 to the left, or in the negative x-direction.

The force between Q_1 and Q_3 is attractive (unlike charges) and pulls Q_1 in the positive y-direction.

We can redraw the diagram as a free-body diagram illustrating the forces to make sure we can visualise the situation:

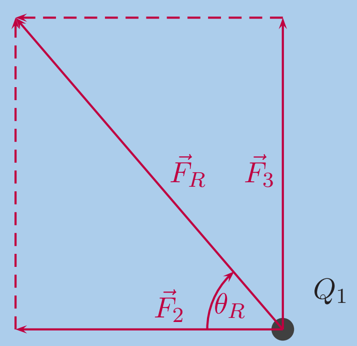


Step 6: Resultant force

The magnitude of the resultant force acting on Q_1 can be calculated from the forces using Pythagoras' theorem because there are only two forces and they act in the x- and y-directions:

$$\begin{aligned} F_R^2 &= F_2^2 + F_3^2 \text{ by Pythagoras' theorem} \\ F_R &= \sqrt{(8{,}630 \times 10^{-5})^2 + (1{,}199 \times 10^{-4})^2} \\ F_R &= 1{,}48 \times 10^{-4}\ \text{N} \end{aligned}$$

and the angle, θ_R made with the x-axis can be found using trigonometry.



$$\tan(\theta_R) = \frac{\text{y-component}}{\text{x-component}}$$

$$\tan(\theta_R) = \frac{1{,}199 \times 10^{-4}}{8{,}630 \times 10^{-5}}$$

$$\theta_R = \tan^{-1}(\frac{1{,}199 \times 10^{-4}}{8{,}630 \times 10^{-5}})$$

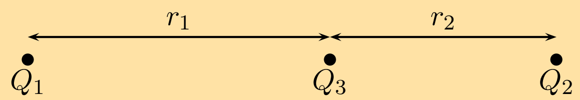
$$\theta_R = 54{,}25° \text{ to 2 decimal places}$$

The final resultant force acting on Q_1 is $1{,}48 \times 10^{-4}$ N acting at an angle of 54,25° to the negative x-axis or 125,75°to the positive x-axis.

We mentioned in grade 10 that charge placed on a spherical conductor spreads evenly along the surface. As a result, if we are far enough from the charged sphere, electrostatically, it behaves as a point-like charge. Thus we can treat spherical conductors (e.g. metallic balls) as point-like charges, with all the charge acting at the centre.

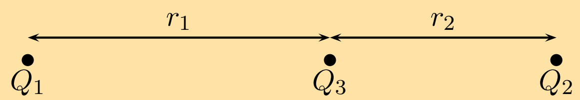
Exercise 9 – 1: Electrostatic forces

1. Calculate the electrostatic force between two charges of +6 nC and +1 nC if they are separated by a distance of 2 mm.
2. What is the magnitude of the repulsive force between two pith balls (a pith ball is a small, light ball that can easily be charged) that are 8 cm apart and have equal charges of −30 nC?
3. How strong is the attractive force between a glass rod with a 0,7 μC charge and a silk cloth with a −0,6 μC charge, which are 12 cm apart, using the approximation that they act like point charges?
4. Two point charges exert a 5 N force on each other. What will the resulting force be if the distance between them is increased by a factor of three?
5. Two point charges are brought closer together, increasing the force between them by a factor of 25. By what factor was their separation decreased?
6. If two equal charges each of 1 C each are separated in air by a distance of 1 km, what is the magnitude of the force acting between them?
7. Calculate the distance between two charges of +4 nC and −3 nC if the electrostatic force between them is 0,005 N.
8. For the charge configuration shown, calculate the resultant force on Q_2 if:
   - $Q_1 = 2{,}3 \times 10^{-7}$ C
   - $Q_2 = 4 \times 10^{-6}$ C
   - $Q_3 = 3{,}3 \times 10^{-7}$ C
   - $r_1 = 2{,}5 \times 10^{-1}$ m
   - $r_2 = 3{,}7 \times 10^{-2}$ m

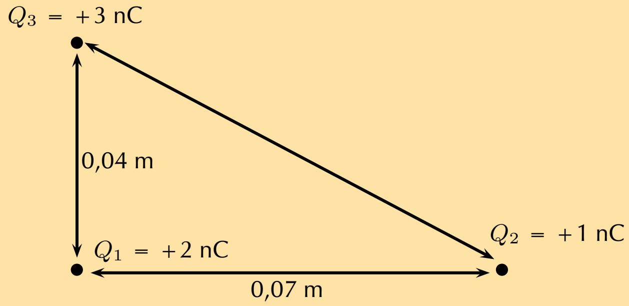


9. For the charge configuration shown, calculate the charge on Q_3 if the resultant force on Q_2 is $6{,}3 \times 10^{-1}$ N to the right and:

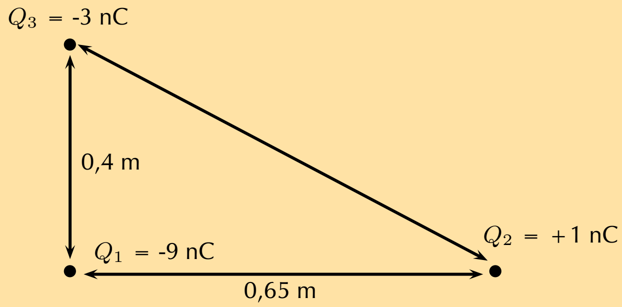
- $Q_1 = 4{,}36 \times 10^{-6}$ C
- $Q_2 = -7 \times 10^{-7}$ C
- $r_1 = 1{,}85 \times 10^{-1}$ m
- $r_2 = 4{,}7 \times 10^{-2}$ m



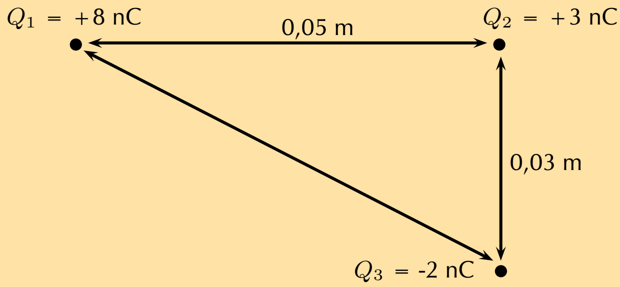
10. Calculate the resultant force on Q_1 given this charge configuration:



11. Calculate the resultant force on Q_1 given this charge configuration:



12. Calculate the resultant force on Q_2 given this charge configuration:



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9.3 Electric field ESBPK

We have seen in the previous section that point charges exert forces on each other even when they are far apart and not touching each other. How do the charges 'know' about the existence of other charges around them?

The answer is that you can think of every charge as being surrounded in space by an electric field. The electric field is the region of space in which an electric charge will experience a force. The direction of the electric field represents the direction of the force a positive test charge would experience if placed in the electric field. In other words, the direction of an electric field at a point in space is the same direction in which a positive test charge would move if placed at that point.

DEFINITION: *Electric field*

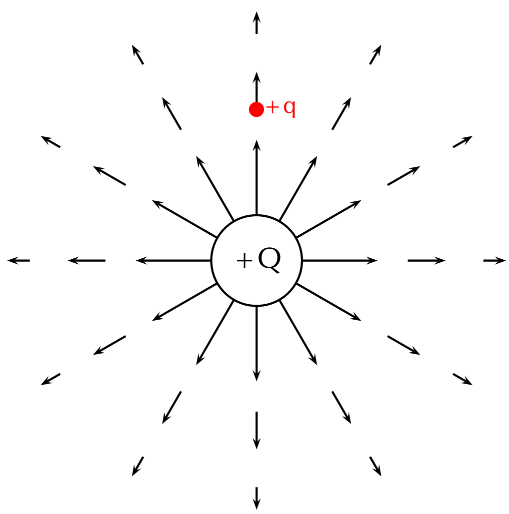
A region of space in which an electric charge will experience a force. The direction of the field at a point in space is the direction in which a positive test charge would moved if placed at that point.

Representing electric fields ESBPM

We can represent the strength and direction of an electric field at a point using **electric field lines**. This is similar to representing magnetic fields around magnets using magnetic field lines as you studied in Grade 10. In the following we will study what the electric fields look like around isolated charges.

Positive charge acting on a test charge

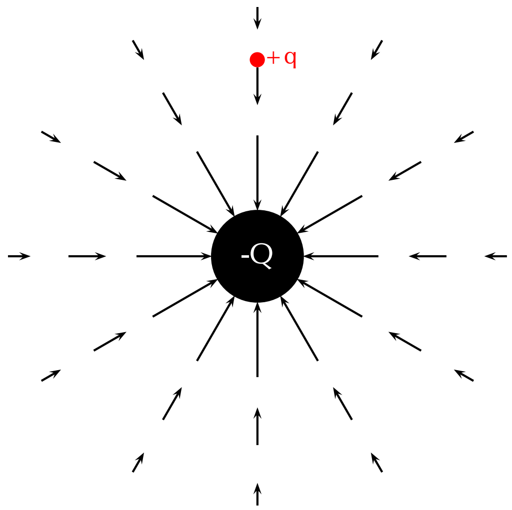
The magnitude of the force that a test charge experiences due to another charge is governed by Coulomb's law. In the diagram below, at each point around the positive charge, $+Q$, we calculate the force a positive test charge, $+q$, would experience, and represent this force (a vector) with an arrow. The force vectors for some points around $+Q$ are shown in the diagram along with the positive test charge $+q$ (in red) located at one of the points.



At every point around the charge $+Q$, the positive test charge, $+q$, will experience a force pushing it away. This is because both charges are positive and so they repel each other. We cannot draw an arrow at every point but we include enough arrows to illustrate what the field would look like. The arrows represent the force the test charge would experience at each point. Coulomb's law is an inverse-square law which means that the force gets weaker the greater the distance between the two charges. This is why the arrows get shorter further away from $+Q$.

Negative charge acting on a test charge

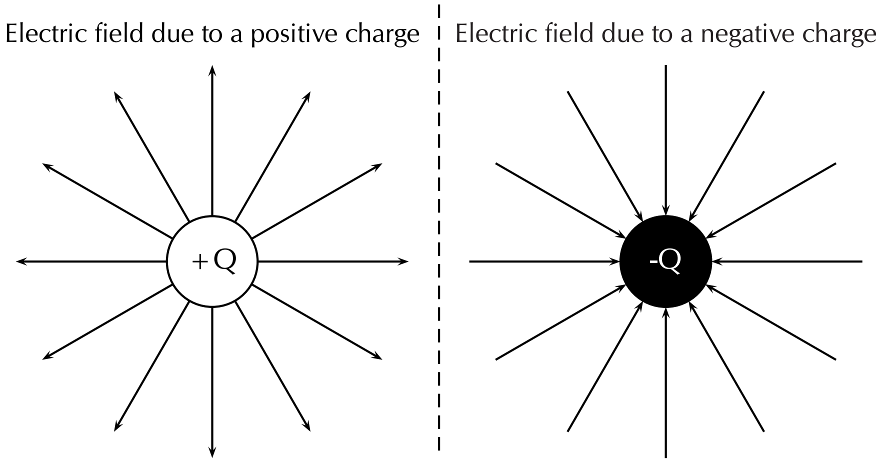
For a negative charge, $-Q$, and a positive test charge, $+q$, the force vectors would look like:



Notice that it is almost identical to the positive charge case. The arrows are the same lengths as in the previous diagram because the absolute magnitude of the charge is the same and so is the magnitude of the test charge. Thus the magnitude of the force is the same at the same points in space. However, the arrows point in the opposite direction because the charges now have opposite signs and attract each other.

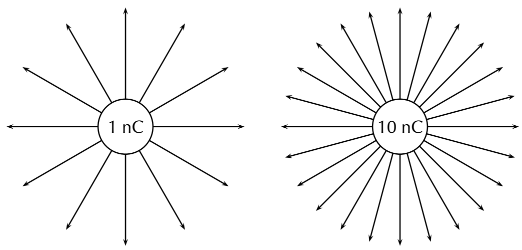
Electric fields around isolated charges - summary

Now, to make things simpler, we draw continuous lines that are tangential to the force that a test charge would experience at each point. The field lines are closer together where the field is stronger. Look at the diagram below: close to the central charges, the field lines are close together. This is where the electric field is strongest. Further away from the central charges where the electric field is weaker, the field lines are more spread out from each other.



We use the following conventions when drawing electric field lines:

- Arrows on the field lines indicate the direction of the field, i.e. the direction in which a positive test charge would move if placed in the field.
- Electric field lines point away from positive charges (like charges repel) and towards negative charges (unlike charges attract).
- Field lines are drawn closer together where the field is stronger.
- Field lines do not touch or cross each other.
- Field lines are drawn perpendicular to a charge or charged surface.
- The greater the magnitude of the charge, the stronger its electric field. We represent this by drawing more field lines around the greater charge than for charges with smaller magnitudes.



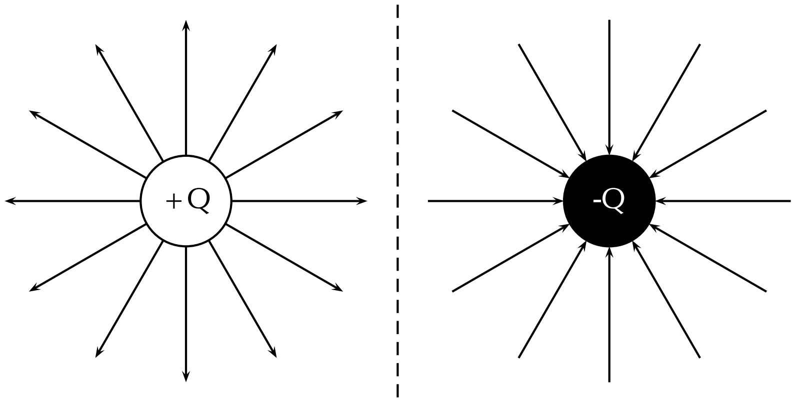
Some important points to remember about electric fields:

- There is an electric field at **every point** in space surrounding a charge.
- Field lines are merely a **representation** – they are not real. When we draw them, we just pick convenient places to indicate the field in space.
- Field lines exist in three dimensions, not only in two dimension as we've drawn them.
- The number of field lines passing through a surface is proportional to the charge contained inside the surface.

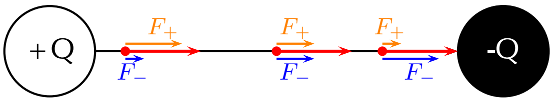
We have seen what the electric fields look like around isolated positive and negative charges. Now we will study what the electric fields look like around combinations of charges placed close together.

Electric field around two unlike charges

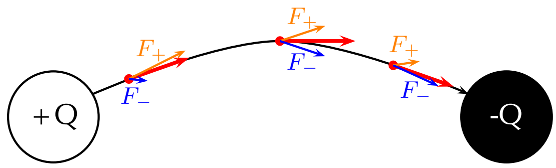
We will start by looking at the electric field around a positive and negative charge placed next to each other. Using the rules for drawing electric field lines, we will sketch the electric field one step at a time. The net resulting field is the sum of the fields from each of the charges. To start off let us sketch the electric fields for each of the charges separately.



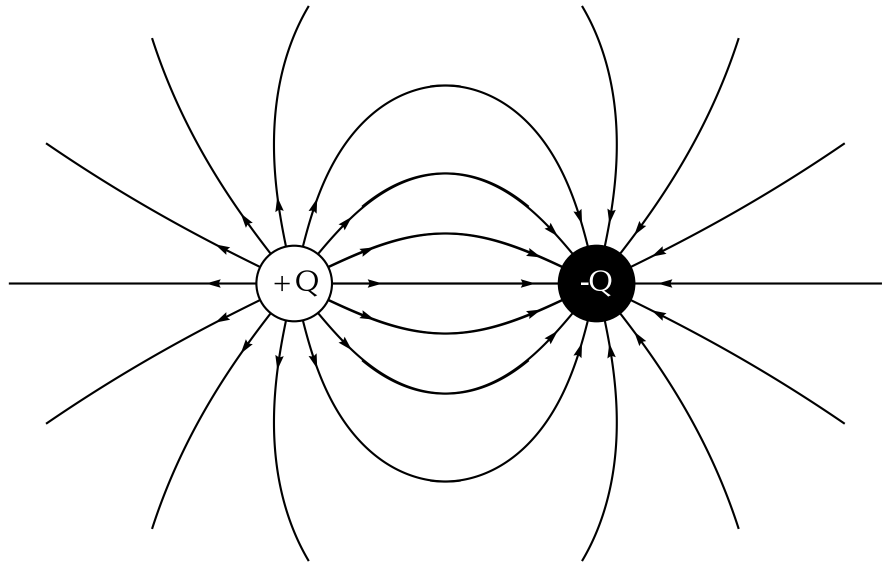
A positive test charge (red dots) placed at different positions directly between the two charges would be pushed away (orange force arrows) from the positive charge and pulled towards (blue force arrows) the negative charge in a straight line. The orange and blue force arrows have been drawn slightly offset from the dots for clarity. In reality they would lie on top of each other. Notice that the further from the positive charge, the smaller the repulsive force, F_+ (shorter orange arrows) and the closer to the negative charge the greater the attractive force, F_- (longer blue arrows). The resultant forces are shown by the red arrows. The electric field line is the black line which is tangential to the resultant forces and is a straight line between the charges pointing from the positive to the negative charge.



Now let's consider a positive test charge placed slightly higher than the line joining the two charges. The test charge will experience a repulsive force (F_+ in orange) from the positive charge and an attractive force (F_- in blue) due to the negative charge. As before, the magnitude of these forces will depend on the distance of the test charge from each of the charges according to Coulomb's law. Starting at a position closer to the positive charge, the test charge will experience a larger repulsive force due to the positive charge and a weaker attractive force from the negative charge. At a position half-way between the positive and negative charges, the magnitudes of the repulsive and attractive forces are the same. If the test charge is placed closer to the negative charge, then the attractive force will be greater and the repulsive force it experiences due to the more distant positive charge will be weaker. At each point we add the forces due to the positive and negative charges to find the resultant force on the test charge (shown by the red arrows). The resulting electric field line, which is tangential to the resultant force vectors, will be a curve.

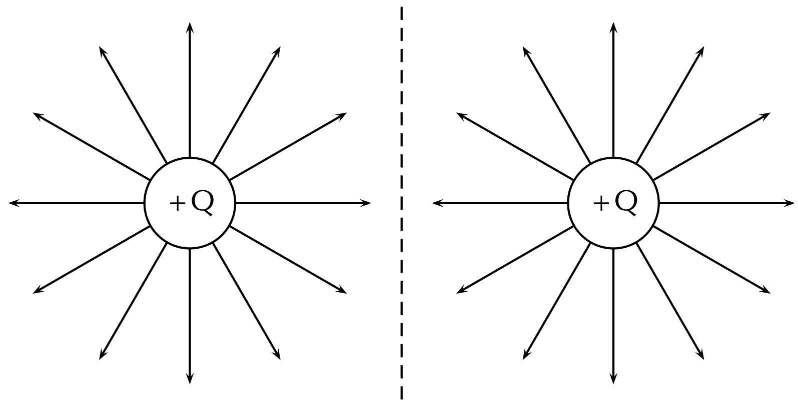


Now we can fill in the other field lines quite easily using the same ideas. The electric field lines look like:

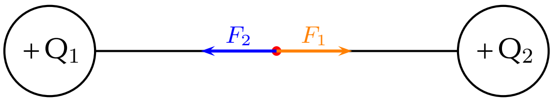


Electric field around two like charges (both positive)

For the case of two positive charges Q_1 and Q_2 of the same magnitude, things look a little different. We can't just turn the arrows around the way we did before. In this case the positive test charge is repelled by both charges. The electric fields around each of the charges in isolation looks like.

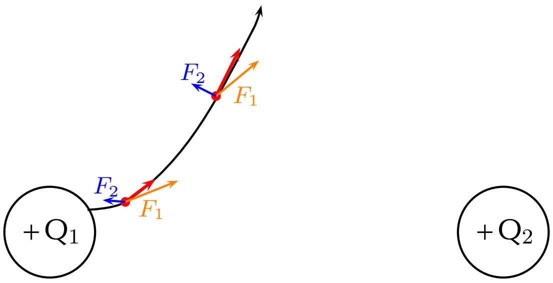


Now we can look at the resulting electric field when the charges are placed next to each other. Let us start by placing a positive test charge directly between the two charges. We can draw the forces exerted on the test charge due to Q_1 and Q_2 and determine the resultant force.

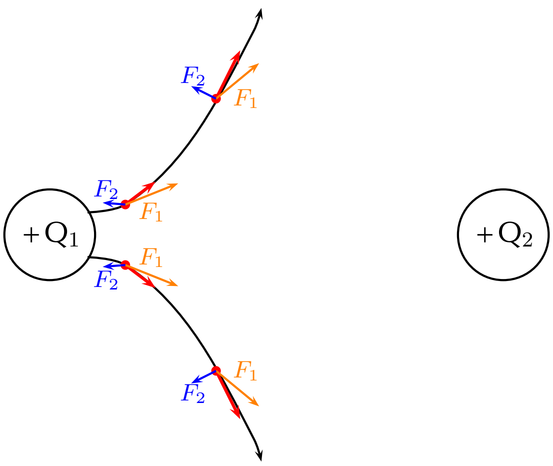


The force F_1 (in orange) on the test charge (red dot) due to the charge Q_1 is equal in magnitude but opposite in direction to F_2 (in blue) which is the force exerted on the test charge due to Q_2. Therefore they cancel each other out and there is no resultant force. This means that the electric field directly between the charges cancels out in the middle. A test charge placed at this point would not experience a force.

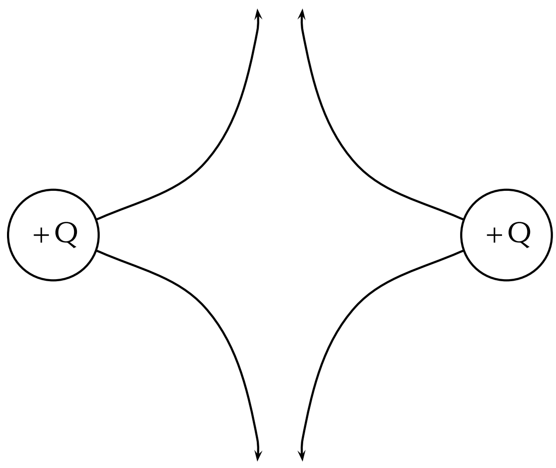
Now let's consider a positive test charge placed close to Q_1 and above the imaginary line joining the centres of the charges. Again we can draw the forces exerted on the test charge due to Q_1 and Q_2 and sum them to find the resultant force (shown in red). This tells us the direction of the electric field line at each point. The electric field line (black line) is tangential to the resultant forces.



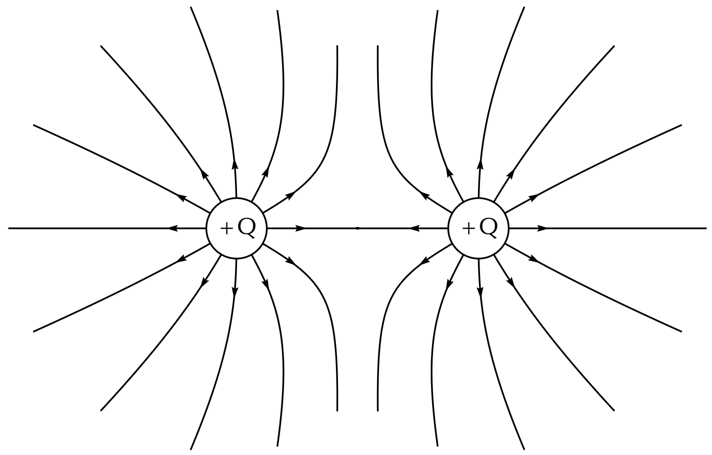
If we place a test charge in the same relative positions but *below* the imaginary line joining the centres of the charges, we can see in the diagram below that the resultant forces are reflections of the forces above. Therefore, the electric field line is just a reflection of the field line above.



Since Q_2 has the same charge as Q_1, the forces at the same relative points close to Q_2 will have the same magnitudes but opposite directions i.e. they are also reflections . We can therefore easily draw the next two field lines as follows:

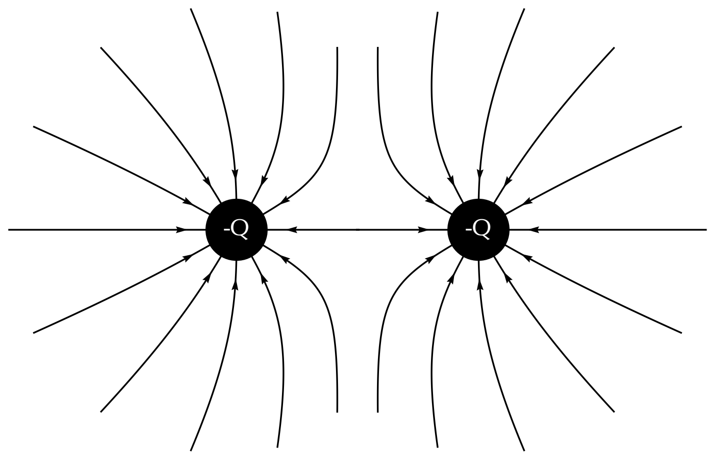


Working through a number of possible starting points for the test charge we can show the electric field can be represented by:



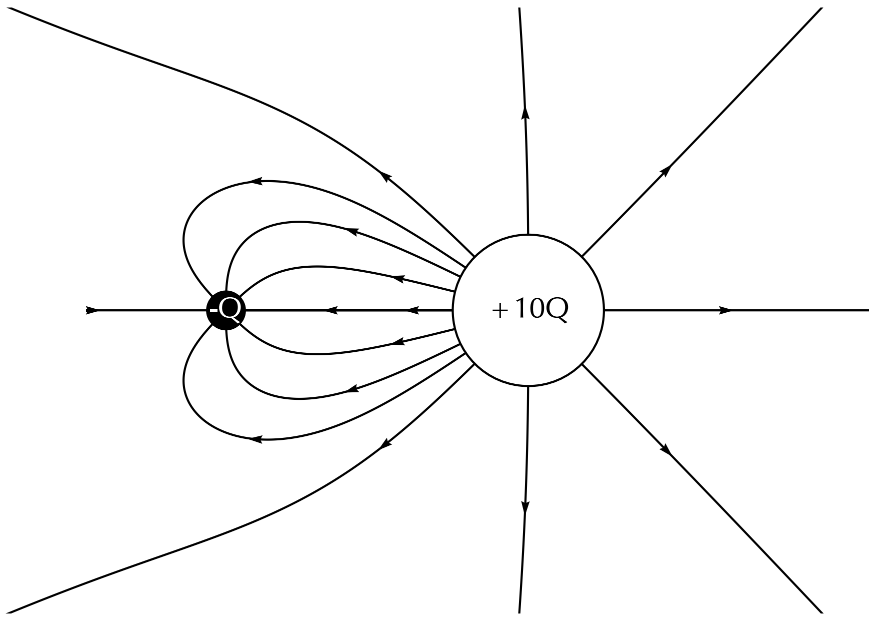
Electric field around two like charges (both negative)

We can use the fact that the direction of the force is reversed for a test charge if you change the sign of the charge that is influencing it. If we change to the case where both charges are negative we get the following result:



Charges of different magnitudes

When the magnitudes are not equal the larger charge will influence the direction of the field lines more than if they were equal. For example, here is a configuration where the positive charge is much larger than the negative charge. You can see that the field lines look more similar to that of an isolated charge at greater distances than in the earlier example. This is because the larger charge gives rise to a stronger field and therefore makes a larger relative contribution to the force on a test charge than the smaller charge.



Electric field strength

ESBPP

In the previous sections we have studied how we can represent the electric fields around a charge or combination of charges by means of electric field lines. In this representation we see that the electric field strength is represented by how close together the field lines are. In addition to the drawings of the electric field, we would also like to be able to quantify (put a number to) how strong an electric field is and what its direction is at any point in space.

A small test charge q placed near a charge Q will experience a force due to the electric field surrounding Q. The magnitude of the force is described by Coulomb's law and depends on the magnitude of the charge Q and the distance of the test charge from Q. The closer the test charge q is to the charge Q, the greater the force it will experience. Also, at points closer to the charge Q, the stronger is its electric field. We define the electric field at a point as the force per unit charge.

DEFINITION: *Electric field*

The magnitude of the electric field, E, at a point can be quantified as the force per unit charge We can write this as:

$$E = \frac{F}{q}$$

where F is the Coulomb force exerted by a charge on a test charge q.
The units of the electric field are newtons per coulomb: N·C^{-1}.

Since the force F is a vector and q is a scalar, the electric field, E, is also a vector; it has a magnitude and a direction at every point.

Given the definition of electric field above and substituting the expression for Coulomb's law for F:

$$\begin{aligned} E &= \frac{F}{q} \\ &= \frac{kQq}{r^2q} \\ E &= \frac{kQ}{r^2} \end{aligned}$$

we can see that the electric field E only depends on the charge Q and not the magnitude of the test charge.

If the electric field is known, then the electrostatic force on any charge q placed into the field is simply obtained by rearranging the definition equation:

$$F = qE.$$

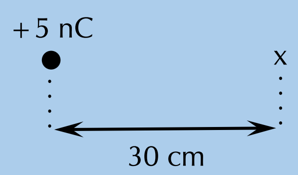
See video: 23Z6 at www.everythingscience.co.za

See simulation: 23Z7 at www.everythingscience.co.za

Worked example 5: Electric field 1

QUESTION

Calculate the electric field strength 30 cm from a 5 nC charge.



SOLUTION

Step 1: Determine what is required

We need to calculate the electric field a distance from a given charge.

Step 2: Determine what is given

We are given the magnitude of the charge and the distance from the charge.

Step 3: Determine how to approach the problem

We will use the equation: $E = k\frac{Q}{r^2}$.

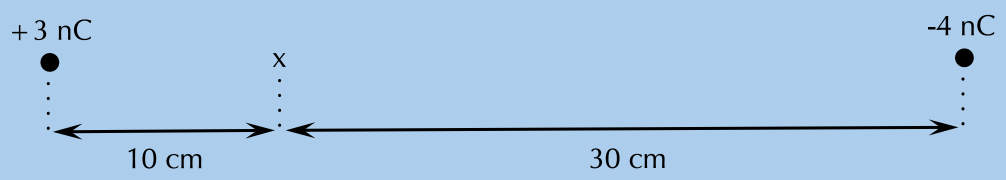
Step 4: Solve the problem

$$E = \frac{kQ}{r^2}$$
$$= \frac{(9{,}0 \times 10^9)(5 \times 10^{-9})}{0{,}3^2}$$
$$= 4{,}99 \times 10^2 \text{ N·C}^{-1}$$

Worked example 6: Electric field 2

QUESTION

Two charges of $Q_1 = +3\text{nC}$ and $Q_2 = -4\text{nC}$ are separated by a distance of 50 cm. What is the electric field strength at a point that is 20 cm from Q_1 and 50 cm from Q_2? The point lies between Q_1 and Q_2.



SOLUTION

Step 1: Determine what is required

We need to calculate the electric field a distance from two given charges.

Step 2: Determine what is given

We are given the magnitude of the charges and the distances from the charges.

Step 3: Determine how to approach the problem

We will use the equation: $E = k\frac{Q}{r^2}$.

We need to calculate the electric field for each charge separately and then add them to determine the resultant field.

Step 4: Solve the problem

We first solve for Q_1:

$$E = \frac{kQ}{r^2}$$
$$= \frac{(9{,}0 \times 10^9)(3 \times 10^{-9})}{0{,}2^2}$$
$$= 6{,}74 \times 10^2 \text{ N·C}^{-1}$$

Then for Q_2:

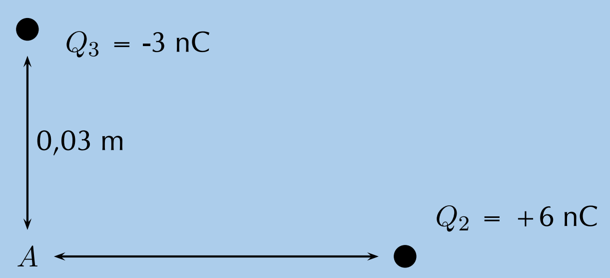
$$E = \frac{kQ}{r^2}$$
$$= \frac{(9{,}0 \times 10^9)(4 \times 10^{-9})}{0{,}3^2}$$
$$= 3{,}99 \times 10^2 \text{ N·C}^{-1}$$

We need to add the two electric fields because both are in the same direction. The field is away from Q_1 and towards Q_2. Therefore, $E_{\text{total}} = 6{,}74 \times 10^2 + 3{,}99 \times 10^2 = 1{,}08 \times 10^2\ \text{N·C}^{-1}$

Worked example 7: Two-dimensional electric fields

QUESTION

Two point charges form a right-angled triangle with the point A at the origin. Their charges are $Q_2 = 6 \times 10^{-9}$ C $= 6$ nC and $Q_3 = -3 \times 10^{-9}$ C $= -3$ nC. The distance between A and Q_2 is 5×10^{-2} m and the distance between A and Q_3 is 3×10^{-2} m. What is the net electric field measured at A from the two charges if they are arranged as shown?



SOLUTION

Step 1: Determine what is required

We are required to calculate the net electric field at A. This field is the sum of the two electric fields - the field from Q_2 at A and from Q_3 at A.

Step 2: Determine how to approach the problem

- We need to calculate the two fields at A, using $E = k\frac{Q}{r^2}$ for the magnitude and determining the direction from the charge signs.
- We then need to add up the two fields using our rules for adding vector quantities, because the electric field is a vector quantity.

Step 3: Determine what is given

We are given all the charges and the distances.

Step 4: Calculate the magnitude of the fields.

The magnitude of the field from Q_2 at A, which we will call E_2, is:

$$
\begin{aligned}
E_2 &= k\frac{Q_2}{r^2} \\
&= (9{,}0 \times 10^9)\frac{(6 \times 10^{-9})}{(5 \times 10^{-2})^2} \\
&= (9{,}0 \times 10^9)\frac{(6 \times 10^{-9})}{(25 \times 10^{-4})} \\
&= 2{,}158 \times 10^4 \text{ N·C}^{-1}
\end{aligned}
$$

The magnitude of the electric field from Q_3 at A, which we will call E_3, is:

$$
\begin{aligned}
E_3 &= k\frac{Q_3}{r^2} \\
&= (9{,}0 \times 10^9)\frac{(3 \times 10^{-9})}{(3 \times 10^{-2})^2} \\
&= (9{,}0 \times 10^9)\frac{(3 \times 10^{-9})}{(9 \times 10^{-4})} \\
&= 2{,}997 \times 10^4 \text{ N·C}^{-1}
\end{aligned}
$$

Step 5: Vector addition of electric fields

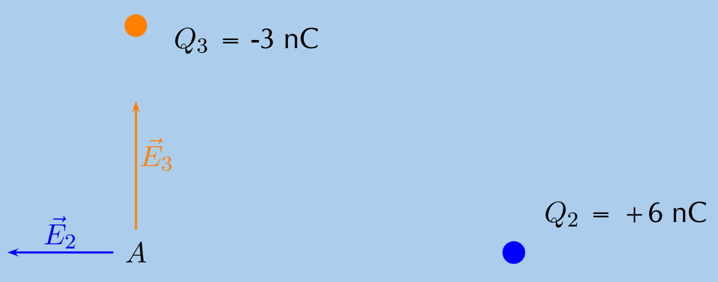
We will use precisely the **same procedure** as before. Determine the vectors on the Cartesian plane, break them into components in the x- and y-directions, sum components in each direction to get the components of the resultant.

We choose the positive directions to be to the right (the positive x-direction) and up (the positive y-direction). We know the electric field magnitudes but we need to use the charges to determine the direction. Then we can use the diagram to determine the directions.

The force between a positive test charge and Q_2 is repulsive (like charges). This means that the electric field is to the left, or in the negative x-direction.

The force between a positive test charge and Q_3 is attractive (unlike charges) and the electric field will be in the positive y-direction.

We can redraw the diagram illustrating the fields to make sure we can visualise the situation:

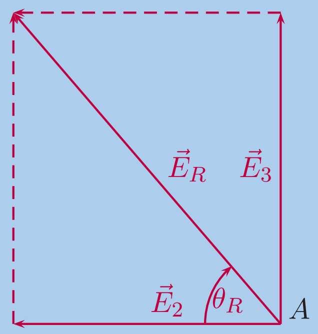


Step 6: Resultant force

The magnitude of the resultant force acting on Q_1 can be calculated from the forces using Pythagoras' theorem because there are only two forces and they act in the x- and y-directions:

$$E_R^2 = E_2^2 + E_3^2 \text{ Pythagoras' theorem}$$
$$E_R = \sqrt{(2{,}158 \times 10^4)^2 + (2{,}997 \times 10^4)^2}$$
$$E_R = 3{,}693 \times 10^4 \text{ N·C}^{-1}$$

and the angle, θ_R made with the x-axis can be found using trigonometry.

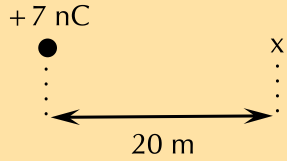


$$\tan(\theta_R) = \frac{\text{y-component}}{\text{x-component}}$$
$$\tan(\theta_R) = \frac{2{,}997 \times 10^4}{2{,}158 \times 10^4}$$
$$\theta_R = \tan^{-1}(\frac{2{,}997 \times 10^4}{2{,}158 \times 10^4})$$
$$\theta_R = 54{,}24°$$

The final resultant electric field acting at A is $3{,}693 \times 10^4$ N·C^{-1} acting at 54,24° to the negative x-axis or 125,76°to the positive x-axis.

Exercise 9 – 2: Electric fields

1. Calculate the electric field strength 20 m from a 7 nC charge.



2. Two charges of $Q_1 = -6$ pC and $Q_2 = -8$ pC are separated by a distance of 3 km. What is the electric field strength at a point that is 2 km from Q_1 and 1 km from Q_2? The point lies between Q_1 and Q_2.

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1. 23Z8 2. 23Z9

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9.4 Chapter summary ESBPQ

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- Objects can be **positively**, **negatively** charged or **neutral**.
- Coulomb's law describes the electrostatic force between two point charges and can be stated as: the magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.
  $F = \frac{kQ_1Q_2}{r^2}$
- An electric field is a region of space in which an electric charge will experience a force. The direction of the field at a point in space is the direction in which a positive test charge would moved if placed at that point.
- We can represent the electric field using field lines. By convention electric field lines point away from positive charges (like charges repel) and towards negative charges (unlike charges attract).
- The magnitude of the electric field, E, at a point can be quantified as the force per unit charge We can write this as: $E = \frac{F}{q}$ where F is the Coulomb force exerted by a charge on a test charge q. The units of the electric field are newtons per coulomb: N·C^{-1} .
- The electric field due to a point charge Q is defined as the force per unit charge:
  $E = \frac{F}{q} = \frac{kQ}{r^2}$
- The electrostatic force is attractive for unlike charges and repulsive for like charges.

Physical Quantities		
Quantity	**Unit name**	**Unit symbol**
Charge (q)	coulomb	C
Distance (d)	metre	m
Electric field (E)	newtons per coulomb	N·C^{-1}
Force (F)	newton	N

Exercise 9 – 3:

1. Two charges of +3 nC and −5 nC are separated by a distance of 40 cm. What is the electrostatic force between the two charges?
2. Two conducting metal spheres carrying charges of +6 nC and −10 nC are separated by a distance of 20 mm.
   a) What is the electrostatic force between the spheres?
   b) The two spheres are touched and then separated by a distance of 60 mm. What are the new charges on the spheres?
   c) What is new electrostatic force between the spheres at this distance?

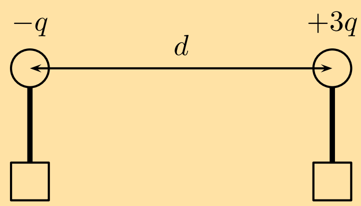
3. The electrostatic force between two charged spheres of +3 nC and +4 nC respectively is 0,4 N. What is the distance between the spheres?

4. Draw the electric field pattern lines between:

   a) two equal positive point charges.

   b) two equal negative point charges.

5. Two small identical metal spheres, on insulated stands, carry charges $-q$ and $+3q$ respectively. When the centres of the spheres are separated by a distance d the one exerts an electrostatic force of magnitude F on the other.

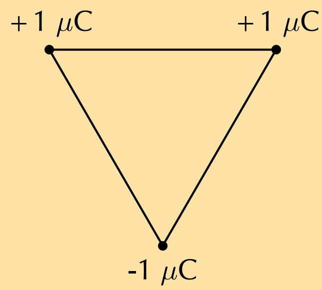


The spheres are now made to touch each other and are then brought back to the same distance d apart. What will be the magnitude of the electrostatic force which one sphere now exerts on the other?

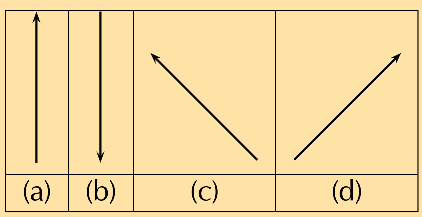
   a) $\frac{1}{4}F$
   b) $\frac{1}{3}F$
   c) $\frac{1}{2}F$
   d) $3F$

[SC 2003/11]

6. Three point charges of magnitude +1 C, +1 C and −1 C respectively are placed on the three corners of an equilateral triangle as shown.



Which vector best represents the direction of the resultant force acting on the −1 C charge as a result of the forces exerted by the other two charges?



[SC 2003/11]

7. a) Write a statement of Coulomb's law.

   b) Calculate the magnitude of the force exerted by a point charge of $+2$ nC on another point charge of -3 nC separated by a distance of 60 mm.

   c) Sketch the electric field between two point charges of $+2$ nC and -3 nC, respectively, placed 60 mm apart from each other.

   [IEB 2003/11 HG1 - Force Fields]

8. The electric field strength at a distance x from a point charge is E. What is the magnitude of the electric field strength at a distance $2x$ away from the point charge?

   a) $\frac{1}{4}E$
   b) $\frac{1}{2}E$
   c) $2E$
   d) $4E$

   [SC 2002/03 HG1]

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1. 23ZC 2. 23ZD 3. 23ZF 4a. 23ZG 4b. 23ZH 5. 23ZJ
6. 23ZK 7. 23ZM 8. 23ZN

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CHAPTER 10

Electromagnetism

10.1 Introduction ESBPR

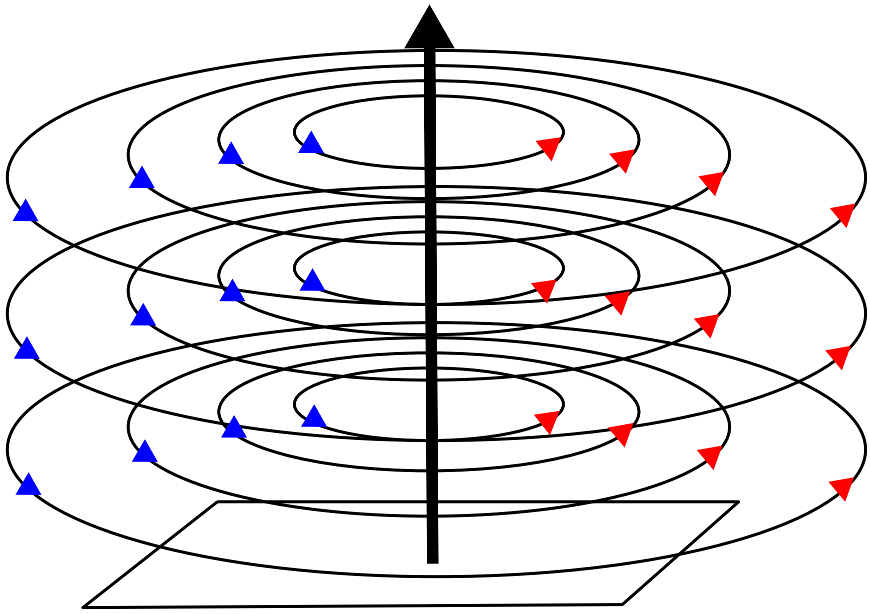
Electromagnetism describes the interaction between charges, currents and the electric and magnetic fields to which they give rise. An electric current creates a magnetic field and a changing magnetic field will create a flow of charge. This relationship between electricity and magnetism has been studied extensively. This has resulted in the invention of many devices which are useful to humans, for example cellular telephones, microwave ovens, radios, televisions and many more.

10.2 Magnetic field associated with a current ESBPS

If you hold a compass near a wire through which current is flowing, the needle on the compass will be deflected.

Since compasses work by pointing along magnetic field lines, this means that there must be a magnetic field near the wire through which the current is flowing.

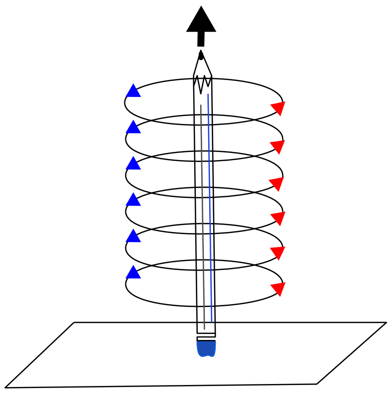
The magnetic field produced by an electric current is always oriented perpendicular to the direction of the current flow. Below is a sketch of what the magnetic field around a wire looks like when the wire has a current flowing in it. We use $\vec{B}$ to denote a magnetic field and arrows on field lines to show the direction of the magnetic field. **Note** that if there is no current there will be no magnetic field.



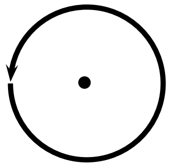
The direction of the current in the conductor (wire) is shown by the central arrow. The circles are field lines and they also have a direction indicated by the arrows on the lines. Similar to the situation with electric field lines, the greater the number of lines (or the closer they are together) in an area the stronger the magnetic field.

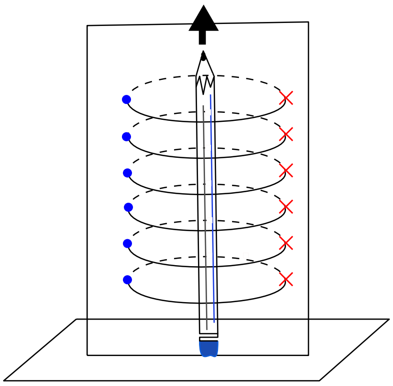
Important: all of our discussion regarding field directions assumes that we are dealing with **conventional current**.

To help you visualise this situation, stand a pen or pencil straight up on a desk. The circles are centred around the pencil or pen and would be drawn parallel to the surface of the desk. The tip of the pen or pencil would point in the direction of the current flow.



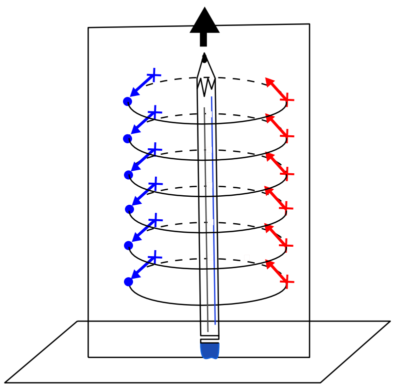
You can look at the pencil or pen from above and the pencil or pen will be a dot in the centre of the circles. The direction of the magnetic field lines is counter-clockwise for this situation.
To make it easier to see what is happening we are only going to draw one set of circular fields lines but note that this is just for the illustration.

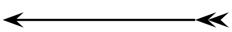




If you put a piece of paper behind the pencil and look at it from the side, then you would be seeing the circular field lines side on and it is hard to know that they are circular. They go through the paper. Remember that field lines have a direction, so when you are looking at the piece of paper from the side it means that the circles go into the paper on one side of the pencil and come out of the paper on the other side.

When we are drawing directions of magnetic fields and currents, we use the symbols $\odot$ and $\otimes$. The symbol $\odot$ represents an arrow that is coming out of the page and the symbol $\otimes$ represents an arrow that is going into the page.
It is easy to remember the meanings of the symbols if you think of an arrow with a sharp tip at the head and a tail with feathers in the shape of a cross.





We will now look at three examples of current carrying wires. For each example we will determine the magnetic field and draw the magnetic field lines around the conductor.

FACT
The Danish physicist, Hans Christian Oersted, was lecturing one day in 1820 on the possibility of electricity and magnetism being related to one another, and in the process demonstrated it conclusively with an experiment in front of his whole class. By passing an electric current through a metal wire suspended above a magnetic compass, Oersted was able to produce a definite motion of the compass needle in response to the current. What began as a guess at the start of the class session was confirmed as fact at the end. Needless to say, Oersted had to revise his lecture notes for future classes. His discovery paved the way for a whole new branch of science - electromagnetism.

The direction of the magnetic field around the current carrying conductor is shown in Figure 10.1.

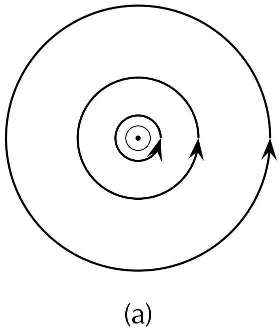


Figure 10.1:

Magnetic field around a conductor when you look at the conductor from one end. (a) Current flows out of the page and the magnetic field is counter-clockwise. (b) Current flows into the page and the magnetic field is clockwise.

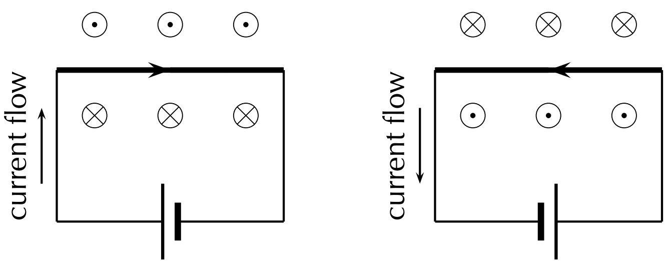


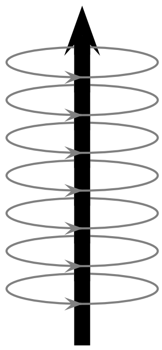
Figure 10.2:

Magnetic fields around a conductor looking down on the conductor. (a) Current flows clockwise. (b) current flows counter-clockwise.

Activity: Direction of a magnetic field

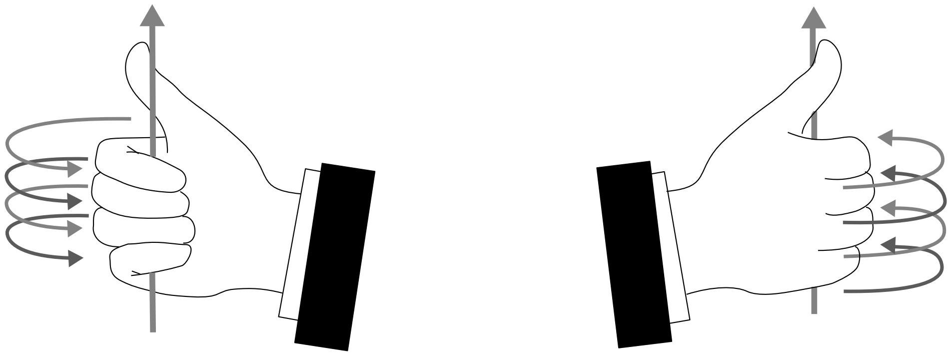
Using the directions given in Figure 10.1 and Figure 10.2 try to find a rule that easily tells you the direction of the magnetic field.

Hint: Use your fingers. Hold the wire in your hands and try to find a link between the direction of your thumb and the direction in which your fingers curl.



The magnetic field around a current carrying conductor.

There is a simple method of finding the relationship between the direction of the current flowing in a conductor and the direction of the magnetic field around the same conductor. The method is called the *Right Hand Rule*. Simply stated, the Right Hand Rule says that the magnetic field lines produced by a current-carrying wire will be oriented in the same direction as the curled fingers of a person's right hand (in the "hitchhiking" position), with the thumb pointing in the direction of the current flow.

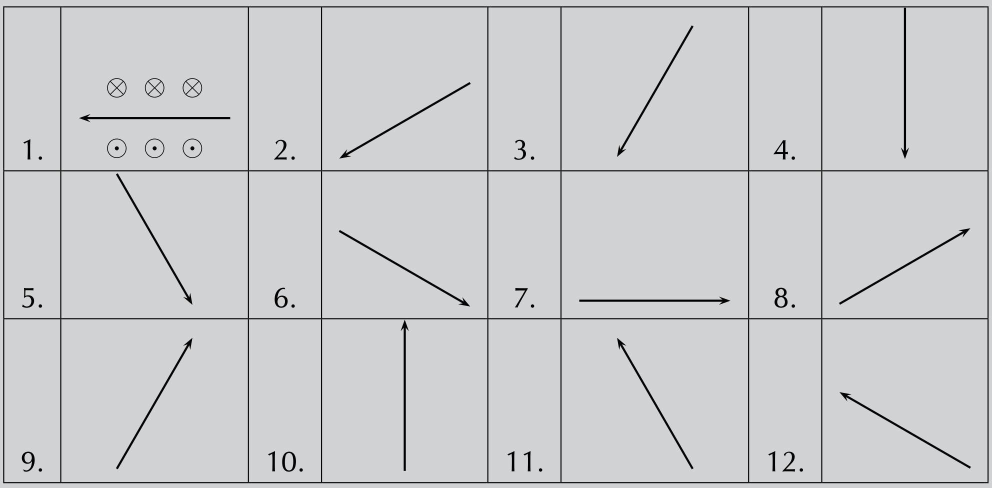


IMPORTANT!

Your right hand and left hand are unique in the sense that you cannot rotate one of them to be in the same position as the other. This means that the right hand part of the rule is essential. You will always get the wrong answer if you use the wrong hand.

Activity: The Right Hand Rule

Use the Right Hand Rule to draw in the directions of the magnetic fields for the following conductors with the currents flowing in the directions shown by the arrows. The first problem has been completed for you.



Activity: Magnetic field around a current carrying conductor

Apparatus:

1. one 9 V battery with holder
2. two hookup wires with alligator clips
3. compass
4. stop watch

Method:

1. Connect your wires to the battery leaving one end of each wire unconnected so that the circuit is not closed.
2. Be sure to limit the current flow to 10 seconds at a time (Why you might ask, the wire has very little resistance on its own so the battery will go flat very quickly). This is to preserve battery life as well as to prevent overheating of the wires and battery contacts.
3. Place the compass close to the wire.
4. Close the circuit and observe what happens to the compass.
5. Reverse the polarity of the battery and close the circuit. Observe what happens to the compass.

Conclusions:

Use your observations to answer the following questions:

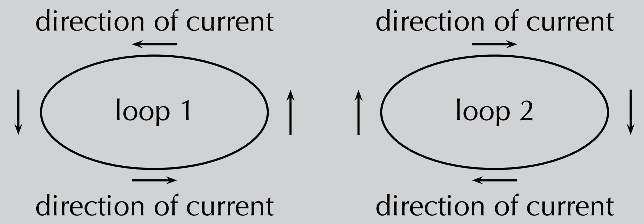
1. Does a current flowing in a wire generate a magnetic field?
2. Is the magnetic field present when the current is not flowing?
3. Does the direction of the magnetic field produced by a current in a wire depend on the direction of the current flow?
4. How does the direction of the current affect the magnetic field?

Magnetic field around a current carrying loop ESBPV

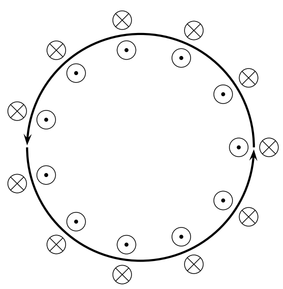
So far we have only looked at straight wires carrying a current and the magnetic fields around them. We are going to study the magnetic field set up by circular loops of wire carrying a current because the field has very useful properties, for example you will see that we can set up a uniform magnetic field.

Activity: Magnetic field around a loop of conductor

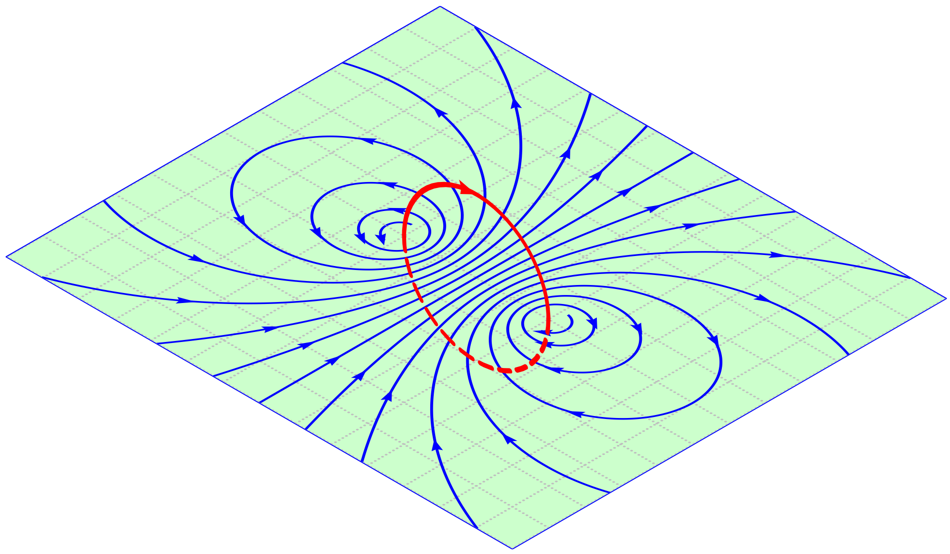
Imagine two loops made from wire which carry currents (in opposite directions) and are parallel to the page of your book. By using the Right Hand Rule, draw what you think the magnetic field would look like at different points around each of the two loops. Loop 1 has the current flowing in a counter-clockwise direction, while loop 2 has the current flowing in a clockwise direction.



If you make a loop of current carrying conductor, then the direction of the magnetic field is obtained by applying the Right Hand Rule to different points in the loop.



The directions of the magnetic field around a loop of current carrying conductor with the current flowing in a counter-clockwise direction is shown.



Notice that there is a variation on the Right Hand Rule. If you make the fingers of your right hand follow the direction of the current in the loop, your thumb will point in the direction where the field lines emerge. This is similar to the north pole (where the field lines emerge from a bar magnet) and shows you which side of the loop would attract a bar magnet's north pole.

Magnetic field around a solenoid ESBPW

If we now add another loop with the current in the same direction, then the magnetic field around each loop can be added together to create a stronger magnetic field. A coil of many such loops is called a *solenoid*. A solenoid is a cylindrical coil of wire acting as a magnet when an electric current flows through the wire. The magnetic field pattern around a solenoid is similar to the magnetic field pattern around the bar magnet that you studied in Grade 10, which had a definite north and south pole as shown in Figure 10.3.

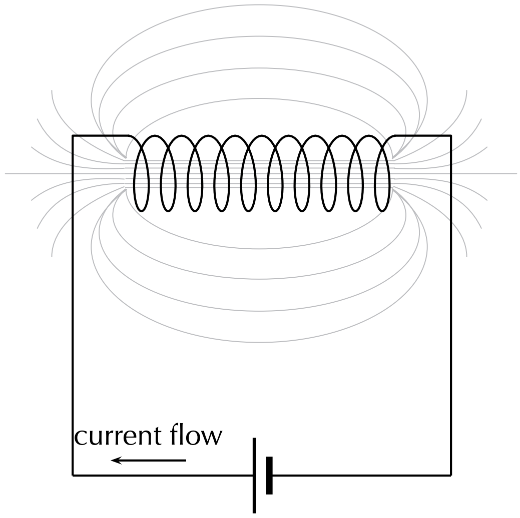
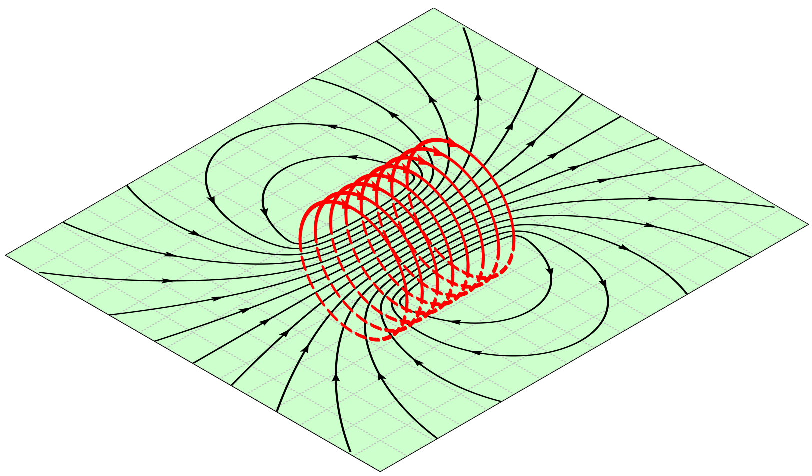


Figure 10.3:

Magnetic field around a solenoid.

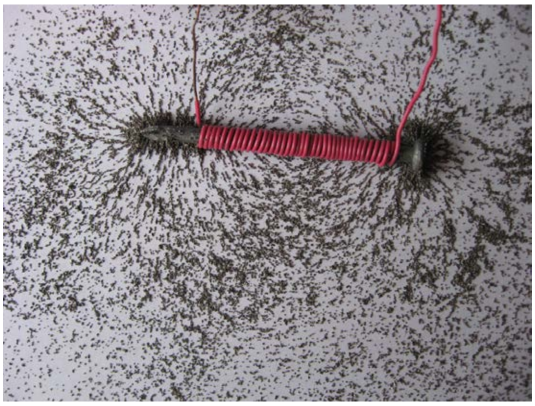


Real-world applications ESBPX

Electromagnets

An *electromagnet* is a piece of wire intended to generate a magnetic field with the passage of electric current through it. Though all current-carrying conductors produce magnetic fields, an electromagnet is usually constructed in such a way as to maximise the strength of the magnetic field it produces for a special purpose. Electromagnets are commonly used in research, industry, medical, and consumer products. An example of a commonly used electromagnet is in security doors, e.g. on shop doors which open automatically.

As an electrically-controllable magnet, electromagnets form part of a wide variety of "electromechanical" devices: machines that produce a mechanical force or motion through electrical power. Perhaps the most obvious example of such a machine is the *electric motor* which will be described in detail in Grade 12. Other examples of the use of electromagnets are electric bells, relays, loudspeakers and scrapyard cranes.



See video: 23ZP at www.everythingscience.co.za

General experiment: Electromagnets

Aim:

A magnetic field is created when an electric current flows through a wire. A single wire does not produce a strong magnetic field, but a wire coiled around an iron core does. We will investigate this behaviour.

Apparatus:

1. a battery and holder
2. a length of wire
3. a compass
4. a few nails

Method:

1. If you have not done the previous experiment in this chapter do it now.
2. Bend the wire into a series of coils before attaching it to the battery. Observe what happens to the deflection of the needle on the compass. Has the deflection of the compass grown stronger?
3. Repeat the experiment by changing the number and size of the coils in the wire. Observe what happens to the deflection on the compass.

4. Coil the wire around an iron nail and then attach the coil to the battery. Observe what happens to the deflection of the compass needle.

Conclusions:

1. Does the number of coils affect the strength of the magnetic field?
2. Does the iron nail increase or decrease the strength of the magnetic field?

Case study: Overhead power lines and the environment

Physical impact:

Power lines are a common sight all across our country. These lines bring power from the power stations to our homes and offices. But these power lines can have negative impacts on the environment. One hazard that they pose is to birds which fly into them. Conservationist Jessica Shaw has spent the last few years looking at this threat. In fact, power lines pose the primary threat to the blue crane, South Africa's national bird, in the Karoo.

"We are lucky in South Africa to have a wide range of bird species, including many large birds like cranes, storks and bustards. Unfortunately, there are also a lot of power lines, which can impact on birds in two ways. They can be electrocuted when they perch on some types of pylons, and can also be killed by colliding with the line if they fly into it, either from the impact with the line or from hitting the ground afterwards.

These collisions often happen to large birds, which are too heavy to avoid a power line if they only see it at the last minute. Other reasons that birds might collide include bad weather, flying in flocks and the lack of experience of younger birds.

Over the past few years we have been researching the serious impact that power line collisions have on Blue Cranes and Ludwig's Bustards. These are two of our endemic species, which means they are only found in southern Africa. They are both big birds that have long lifespans and breed slowly, so the populations might not recover from high mortality rates. We have walked and driven under power lines across the Overberg and the Karoo to count dead birds. The data show that thousands of these birds are killed by collisions every year, and Ludwig's Bustard is now listed as an Endangered species because of this high level of unnatural mortality. We are also looking for ways to reduce this problem, and have been working with Eskom to test different line marking devices. When markers are hung on power lines, birds might be able to see the power line from further away, which will give them enough time to avoid a collision."

Impact of fields:

The fact that a field is created around the power lines means that they can potentially have an impact at a distance. This has been studied and continues to be a topic of significant debate. At the time of writing, the World Health Organisation guidelines for human exposure to electric and magnetic fields indicate that there is no clear link between exposure to the magnetic and electric fields that the general public encounters from power lines, because these are extremely low frequency fields.

Power line noise can interfere with radio communications and broadcasting. Essentially, the power lines or associated hardware improperly generate unwanted radio signals that override or compete with desired radio signals. Power line noise can impact the quality of radio and television reception. Disruption of radio communications, such as amateur radio, can also occur. Loss of critical communications, such as police, fire, military and other similar users of the radio spectrum, can result in even more serious consequences.

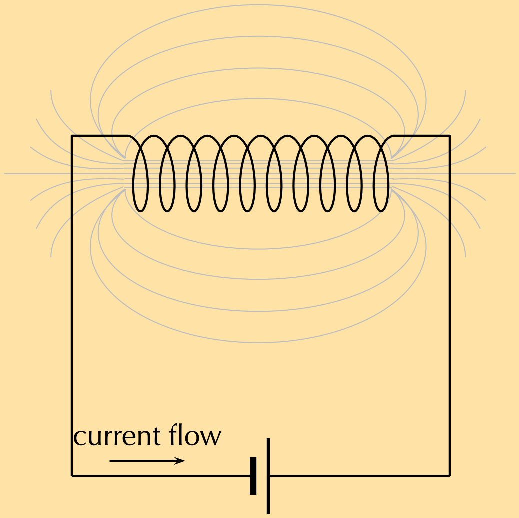
Group discussion:

- Discuss the above information.
- Discuss other ways that power lines affect the environment.

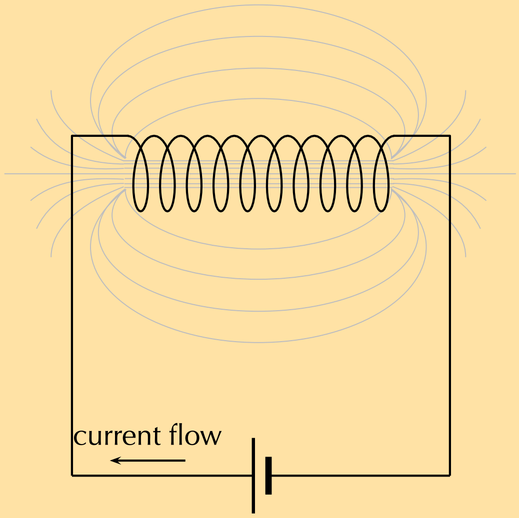
FACT
When lightning strikes a ship or an aeroplane, it can damage or otherwise change its magnetic compass. There have been recorded instances of a lightning strike changing the polarity of the compass so the needle points south instead of north.

Exercise 10 – 1: Magnetic Fields

1. Give evidence for the existence of a magnetic field near a current carrying wire.
2. Describe how you would use your right hand to determine the direction of a magnetic field around a current carrying conductor.
3. Use the Right Hand Rule to determine the direction of the magnetic field for the following situations:

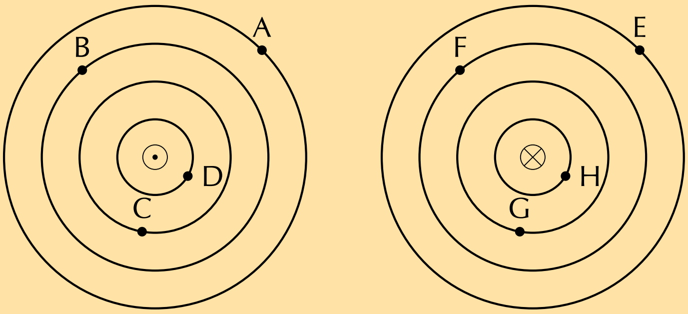


a)



b)

4. Use the Right Hand Rule to find the direction of the magnetic fields at each of the points labelled A - H in the following diagrams.



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1. 23ZQ 2. 23ZR 3a. 23ZS 3b. 23ZT 4. 23ZV

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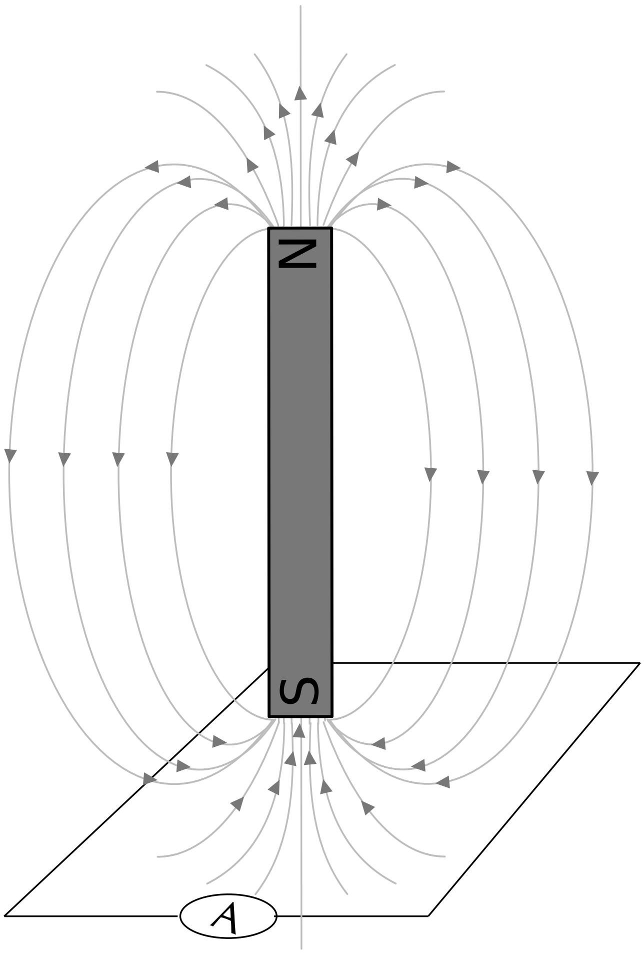
10.3 Faraday's law of electromagnetic induction ESBPY

Current induced by a changing magnetic field ESBPZ

While Oersted's surprising discovery of electromagnetism paved the way for more practical applications of electricity, it was Michael Faraday who gave us the key to the practical generation of electricity: **electromagnetic induction**.

Faraday discovered that when he moved a magnet near a wire a voltage was generated across it. If the magnet was held stationary no voltage was generated, the voltage only existed while the magnet was moving. We call this voltage the induced emf ($\mathcal{E}$).

A circuit loop connected to a sensitive ammeter will register a current if it is set up as in this figure and the magnet is moved up and down:



Magnetic flux

Before we move onto the definition of Faraday's law of electromagnetic induction and examples, we first need to spend some time looking at the magnetic flux. For a loop of area A in the presence of a uniform magnetic field, $\vec{B}$, the magnetic flux (ϕ) is defined as:

$$\phi = BA\cos\theta$$

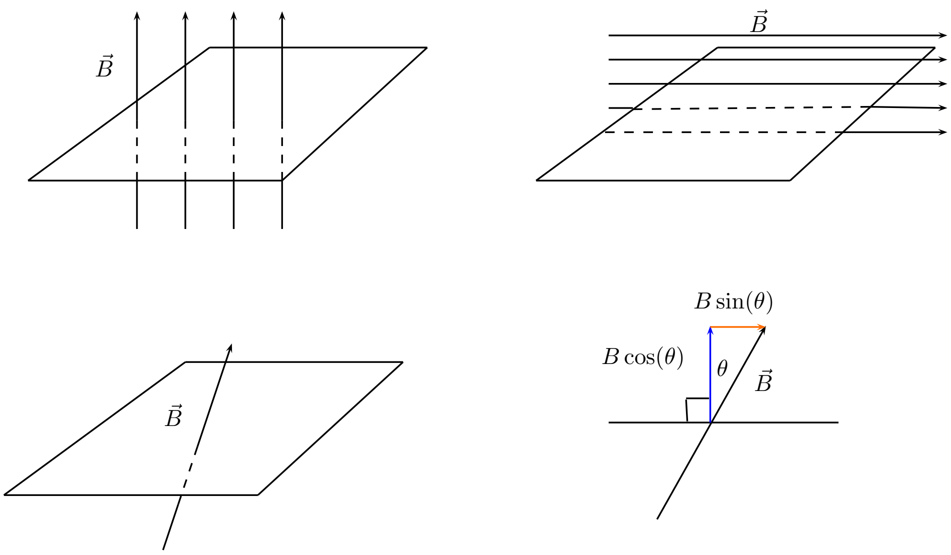
Where:

$\theta =$ the angle between the magnetic field, B, and the normal to the loop of area A
$A =$ the area of the loop
$B =$ the magnetic field

The S.I. unit of magnetic flux is the weber (Wb).

You might ask yourself why the angle θ is included. The flux depends on the magnetic field that passes through surface. We know that a field parallel to the surface can't induce a current because it doesn't pass through the surface. If the magnetic field is not perpendicular to the surface then there is a component which is perpendicular and a component which is parallel to the surface. The parallel component can't contribute to the flux, only the vertical component can.

In this diagram we show that a magnetic field at an angle other than perpendicular can be broken into components. The component perpendicular to the surface has the magnitude $B\cos(\theta)$ where θ is the angle between the normal and the magnetic field.



DEFINITION: *Faraday's Law of electromagnetic induction*

The emf, $\mathcal{E}$, produced around a loop of conductor is proportional to the rate of change of the magnetic flux, ϕ, through the area, A, of the loop. This can be stated mathematically as:

$$\mathcal{E} = -N\frac{\Delta\phi}{\Delta t}$$

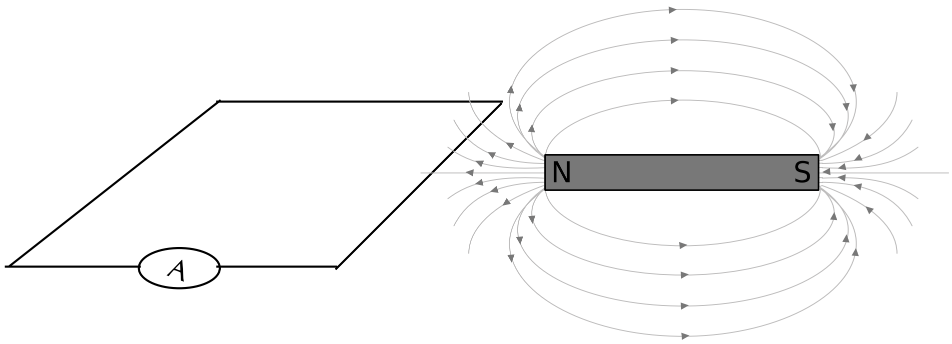
where $\phi = B \cdot A$ and B is the strength of the magnetic field. N is the number of circuit loops. A magnetic field is measured in units of teslas (T). The minus sign indicates direction and that the induced emf tends to oppose the change in the magnetic flux. The minus sign can be ignored when calculating magnitudes.

Faraday's Law relates induced emf to the rate of change of flux, which is the product of the magnetic field and the cross-sectional area through which the field lines pass.

IMPORTANT!

It is not the area of the wire itself but the area that the wire encloses. This means that if you bend the wire into a circle, the area we would use in a flux calculation is the surface area of the circle, not the wire.

In this illustration, where the magnet is in the same plane as the circuit loop, there would be no current even if the magnet were moved closer and further away. This is because the magnetic field lines do not pass through the enclosed area but are parallel to it. The magnetic field lines must pass through the area enclosed by the circuit loop for an emf to be induced.

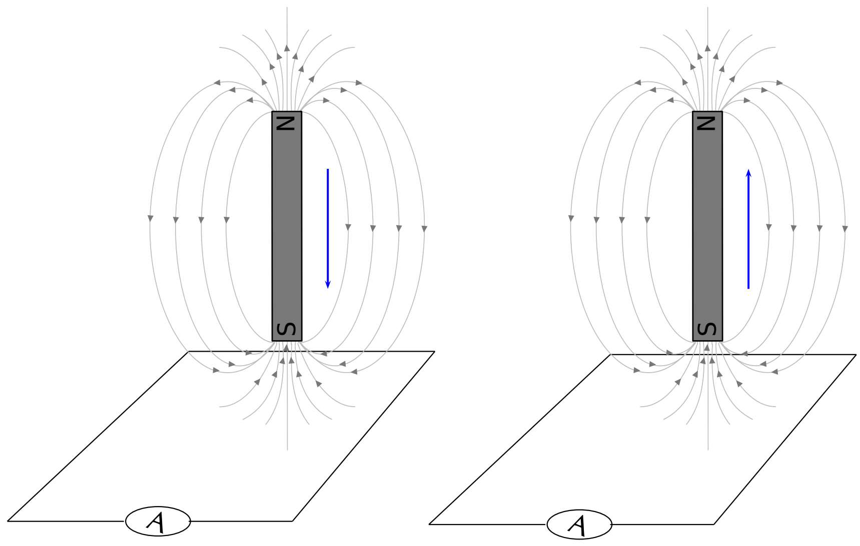


Direction of induced current ESBQ2

The most important thing to remember is that the induced current opposes whatever change is taking place.

In the first picture (left) the circuit loop has the south pole of a magnet moving closer. The magnitude of the field from the magnet is getting larger. The response from the induced emf will be to try to resist the field towards the pole getting stronger. The field is a vector so the current will flow in a direction so that the fields due to the current tend to cancel those from the magnet, keeping the resultant field the same.

To resist the change from an approaching south pole from above, the current must result in field lines that move away from the approaching pole. The induced magnetic field must therefore have field lines that go down on the inside of the loop. The current direction indicated by the arrows on the circuit loop will achieve this. Test this by using the Right Hand Rule. Put your right thumb in the direction of one of the arrows and notice what the field curls downwards into the area enclosed by the loop.



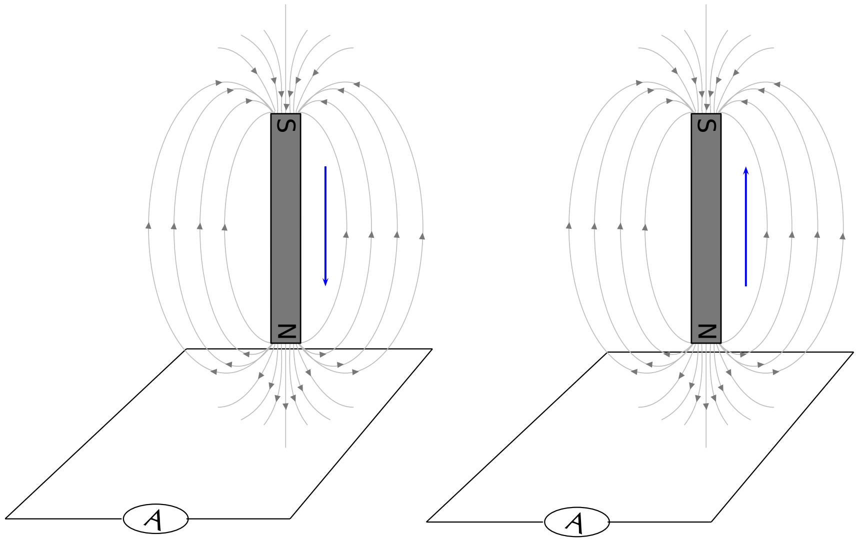
In the second diagram the south pole is moving away. This means that the field from the magnet will be getting weaker. The response from the induced current will be to set up a magnetic field that adds to the existing one from the magnetic to resist it decreasing in strength.

Another way to think of the same feature is just using poles. To resist an approaching south pole the current that is induced creates a field that looks like another south pole on the side of the approaching south pole. Like poles repel, you can think of the current setting up a south pole to repel the approaching south pole. In the second panel, the current sets up a north pole to attract the south pole to stop it moving away.

We can also use the variation of the Right Hand Rule, putting your fingers in the direction of the current to get your thumb to point in the direction of the field lines (or the north pole).

We can test all of these on the cases of a north pole moving closer or further away from the circuit. For the first case of the north pole approaching, the current will resist the change by setting up a field in the opposite direction to the field from the magnet that is getting stronger. Use the Right Hand Rule to confirm that the arrows create a field with field lines that curl upwards in the enclosed area cancelling out those curling downwards from the north pole of the magnet.

Like poles repel, alternatively test that putting the fingers of your right hand in the direction of the current leaves your thumb pointing upwards indicating a north pole.



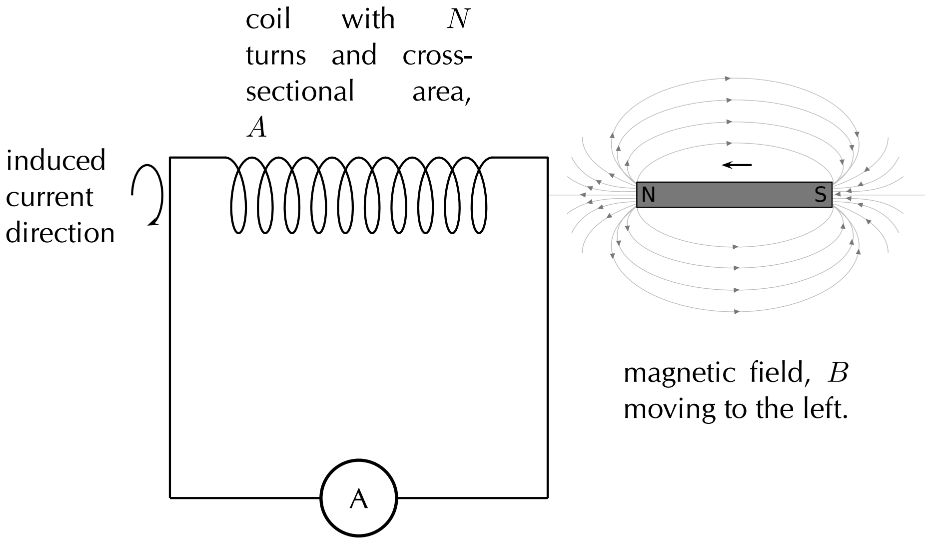
For the second figure where the north pole is moving away the situation is reversed.

Direction of induced current in a solenoid ESBQ3

The approach for looking at the direction of current in a solenoid is the same the approach described above. The only difference being that in a solenoid there are a number of loops of wire so the magnitude of the induced emf will be different. The flux would be calculated using the surface area of the solenoid multiplied by the number of loops.

Remember: the directions of currents and associated magnetic fields can all be found using only the Right Hand Rule. When the fingers of the right hand are pointed in the direction of the magnetic field, the thumb points in the direction of the current. When the thumb is pointed in the direction of the magnetic field, the fingers point in the direction of the current.

The direction of the current will be such as to oppose the change. We would use a setup as in this sketch to do the test:

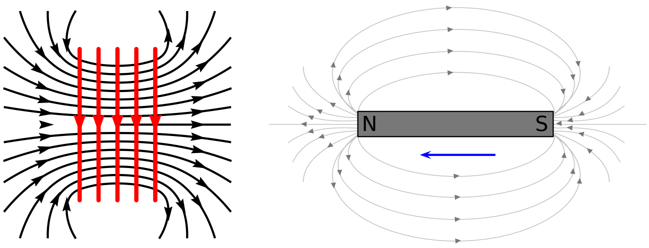


In the case where a north pole is brought towards the solenoid the current will flow so

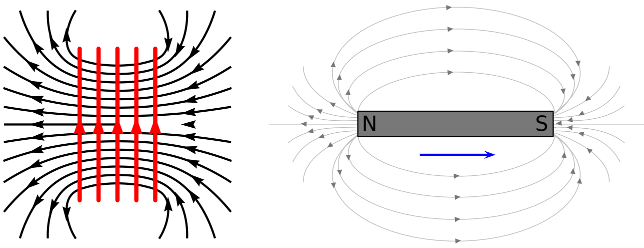
TIP
An easy way to create a magnetic field of changing intensity is to move a permanent magnet next to a wire or coil of wire. The magnetic field must increase or decrease in intensity *perpendicular* to the wire (so that the magnetic field lines "cut across" the conductor), or else no voltage will be induced.

TIP
The induced current generates a magnetic field. The induced magnetic field is in a direction that tends to cancel out the change in the magnetic field in the loop of wire. So, you can use the Right Hand Rule to find the direction of the induced current by remembering that the induced magnetic field is opposite in direction to the change in the magnetic field.

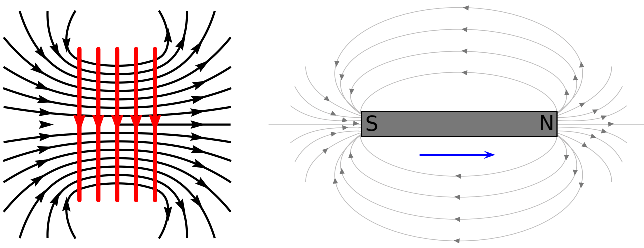
that a north pole is established at the end of the solenoid closest to the approaching magnet to repel it (verify using the Right Hand Rule):



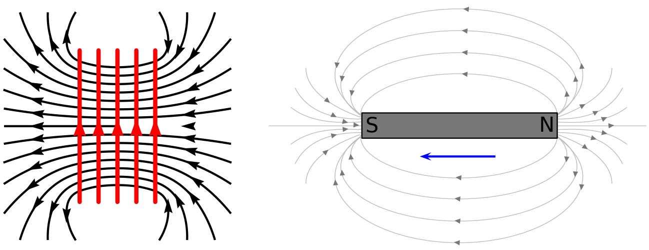
In the case where a north pole is moving away from the solenoid the current will flow so that a south pole is established at the end of the solenoid closest to the receding magnet to attract it:



In the case where a south pole is moving away from the solenoid the current will flow so that a north pole is established at the end of the solenoid closest to the receding magnet to attract it:



In the case where a south pole is brought towards the solenoid the current will flow so that a south pole is established at the end of the solenoid closest to the approaching magnet to repel it:



Induction

Electromagnetic induction is put into practical use in the construction of electrical generators which use mechanical power to move a magnetic field past coils of wire to generate voltage. However, this is by no means the only practical use for this principle.

If we recall, the magnetic field produced by a current-carrying wire is always perpendicular to the wire, and that the flux intensity of this magnetic field varies with the amount of current which passes through it. We can therefore see that a wire is capable of inducing a voltage *along its own length* if the current is changing. This effect is called *self-induction*. Self-induction is when a changing magnetic field is produced by changes in current through a wire, inducing a voltage along the length of that same wire.

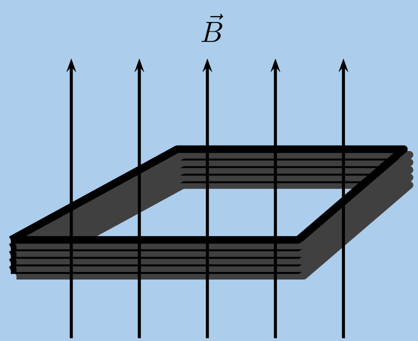
If the magnetic flux is enhanced by bending the wire into the shape of a coil, and/or wrapping that coil around a material of high permeability, this effect of self-induced voltage will be more intense. A device constructed to take advantage of this effect is called an *inductor*.

Remember that the induced current will create a magnetic field that opposes the change in the magnetic flux. This is known as Lenz's law.

Worked example 1: Faraday's law

QUESTION

Consider a flat square coil with 5 turns. The coil is 0,50 m on each side and has a magnetic field of 0,5 T passing through it. The plane of the coil is perpendicular to the magnetic field: the field points out of the page. Use Faraday's Law to calculate the induced emf, if the magnetic field is increases uniformly from 0,5 T to 1 T in 10 s. Determine the direction of the induced current.



SOLUTION

Step 1: Identify what is required

We are required to use Faraday's Law to calculate the induced emf.

Step 2: Write Faraday's Law

$$\mathcal{E} = -N\frac{\Delta\phi}{\Delta t}$$

We know that the magnetic field is at right angles to the surface and so aligned with the normal. This means we do not need to worry about the angle that the field makes with the normal and $\phi = BA$. The starting or initial magnetic field, B_i, is given as is the final field magnitude, B_f. We want to determine the magnitude of the emf so we can ignore the minus sign.

The area, A, is the area of square coil.

Step 3: Solve Problem

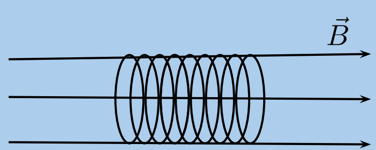
$$\begin{aligned}\mathcal{E} &= N\frac{\Delta\phi}{\Delta t}\\ &= N\frac{\phi_f - \phi_i}{\Delta t}\\ &= N\frac{B_f A - B_i A}{\Delta t}\\ &= N\frac{A(B_f - B_i)}{\Delta t}\\ &= (5)\frac{(0{,}50)^2(1-0{,}50)}{10}\\ &= (5)\frac{(0{,}50)^2(1-0{,}50)}{10}\\ &= 0{,}0625\ \text{V}\end{aligned}$$

The induced current is anti-clockwise as viewed from the direction of the increasing magnetic field.

Worked example 2: Faraday's law

QUESTION

Consider a solenoid of 9 turns with unknown radius, r. The solenoid is subjected to a magnetic field of 0,12 T. The axis of the solenoid is parallel to the magnetic field. When the field is uniformly switched to 12 T over a period of 2 minutes an emf with a magnitude of $-0{,}3$ V is induced. Determine the radius of the solenoid.



SOLUTION

Step 1: Identify what is required

We are required to determine the radius of the solenoid. We know that the relationship between the induced emf and the field is governed by Faraday's law which includes the geometry of the solenoid. We can use this relationship to find the radius.

Step 2: Write Faraday's Law

$$\mathcal{E} = -N\frac{\Delta\phi}{\Delta t}$$

We know that the magnetic field is at right angles to the surface and so aligned with the normal. This means we do not need to worry about the angle the field makes with the normal and $\phi = BA$. The starting or initial magnetic field, B_i, is given as is the final field magnitude, B_f. We can drop the minus sign because we are working with the magnitude of the emf only.

The area, A, is the surface area of the solenoid which is πr^2.

Step 3: Solve Problem

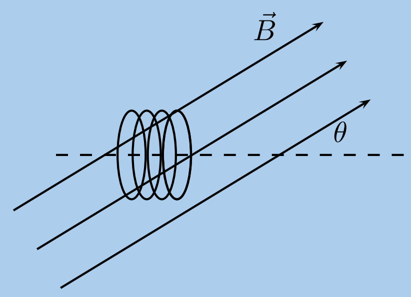
$$\begin{aligned}
\mathcal{E} &= N\frac{\Delta\phi}{\Delta t} \\
&= N\frac{\phi_f - \phi_i}{\Delta t} \\
&= N\frac{B_f A - B_i A}{\Delta t} \\
&= N\frac{A(B_f - B_i)}{\Delta t} \\
(0{,}30) &= (9)\frac{(\pi r^2)(12 - 0{,}12)}{120} \\
r^2 &= \frac{(0{,}30)(120)}{(9)\pi(12 - 0{,}12)} \\
r^2 &= 0{,}107175 \\
r &= 0{,}32 \text{ m}
\end{aligned}$$

The solenoid has a radius of 0,32 m.

Worked example 3: Faraday's law

QUESTION

Consider a circular coil of 4 turns with radius 3×10^{-2} m. The solenoid is subjected to a varying magnetic field that changes uniformly from 0,4 T to 3,4 T in an interval of 27 s. The axis of the solenoid makes an angle of 35° to the magnetic field. Find the induced emf.



SOLUTION

Step 1: Identify what is required

We are required to use Faraday's Law to calculate the induced emf.

Step 2: Write Faraday's Law

$$\mathcal{E} = -N\frac{\Delta\phi}{\Delta t}$$

We know that the magnetic field is at an angle to the surface normal. This means we must account for the angle that the field makes with the normal and $\phi = BA\cos(\theta)$. The starting or initial magnetic field, B_i, is given as is the final field magnitude, B_f. We want to determine the magnitude of the emf so we can ignore the minus sign.

The area, A, will be πr^2.

Step 3: Solve Problem

$$\begin{aligned}
\mathcal{E} &= N\frac{\Delta\phi}{\Delta t} \\
&= N\frac{\phi_f - \phi_i}{\Delta t} \\
&= N\frac{B_f A\cos(\theta) - B_i A\cos(\theta)}{\Delta t} \\
&= N\frac{A\cos(\theta)(B_f - B_i)}{\Delta t} \\
&= (4)\frac{(\pi(0{,}03)^2\cos(35))(3{,}4 - 0{,}4)}{27} \\
&= 1{,}03 \times 10^{-3}\ \text{V}
\end{aligned}$$

The induced current is anti-clockwise as viewed from the direction of the increasing magnetic field.

See simulation: 23ZW at www.everythingscience.co.za

Real-life applications

The following devices use Faraday's Law in their operation.

- induction stoves
- tape players
- metal detectors
- transformers

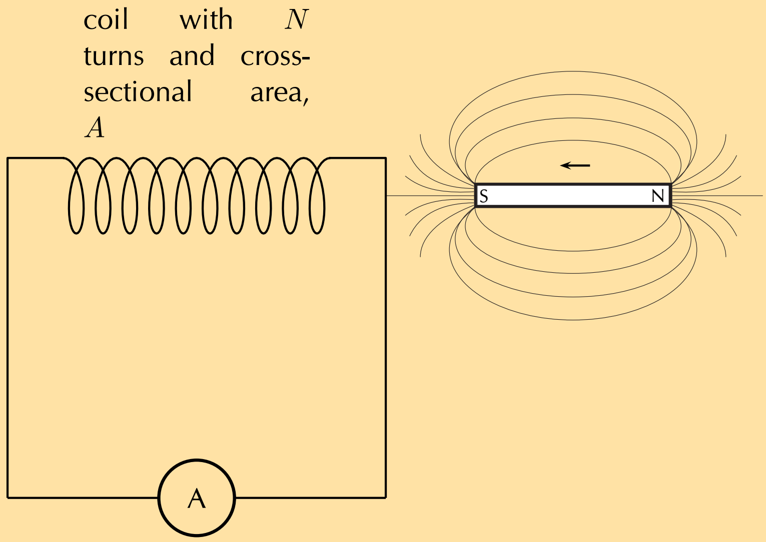
Project: Real-life applications of Faraday's Law

Choose one of the following devices and do some research on the internet, or in a library, how your device works. You will need to refer to Faraday's Law in your explanation.

- induction stoves
- tape players
- metal detectors
- transformers

Exercise 10 – 2: Faraday's Law

1. State Faraday's Law of electromagnetic induction in words and write down a mathematical relationship.
2. Describe what happens when a bar magnet is pushed into or pulled out of a solenoid connected to an ammeter. Draw pictures to support your description.
3. Explain how it is possible for the magnetic flux to be zero when the magnetic field is not zero.
4. Use the Right Hand Rule to determine the direction of the induced current in the solenoid below.



5. Consider a circular coil of 5 turns with radius 1,73 m. The coil is subjected to a varying magnetic field that changes uniformly from 2,18 T to 12,7 T in an interval of 3 minutes. The axis of the solenoid makes an angle of 27° to the magnetic field. Find the induced emf.

6. Consider a solenoid coil of 11 turns with radius $13{,}8 \times 10^{-2}$ m. The solenoid is subjected to a varying magnetic field that changes uniformly from 5,34 T to 2,7 T in an interval of 12 s. The axis of the solenoid makes an angle of 13° to the magnetic field.

   a) Find the induced emf.

   b) If the angle is changed to 67,4°, what would the radius need to be for the emf to remain the same?

7. Consider a solenoid with 5 turns and a radius of 11×10^{-2} m. The axis of the solenoid makes an angle of 23° to the magnetic field.

   a) Find the change in flux if the emf is 12 V over a period of 12 s.

   b) If the angle is changed to 45°, what would the time interval need to change to for the induced emf to remain the same?

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10.4 Chapter summary

ESBQ4

See presentation: 2426 at www.everythingscience.co.za

- Electromagnetism is the study of the properties and relationship between electric currents and magnetism.
- A current-carrying conductor will produce a magnetic field around the conductor.
- The direction of the magnetic field is found by using the Right Hand Rule.
- Electromagnets are temporary magnets formed by current-carrying conductors.
- The magnetic flux through a surface is the product of the component of the magnetic field normal to the surface and the surface area, $\phi = BA\cos(\theta)$.
- Electromagnetic induction occurs when a changing magnetic field induces a voltage in a current-carrying conductor.
- The magnitude of the induced emf is given by Faraday's law of electromagnetic induction: $\mathcal{E} = -N\frac{\Delta\phi}{\Delta t}$

Physical Quantities		
Quantity	**Unit name**	**Unit symbol**
Induced emf ($\mathcal{E}$)	volt	V
Magnetic field (B)	tesla	T
Magnetic flux (ϕ)	weber	Wb
Time (t)	seconds	s

Exercise 10 – 3:

1. What did Hans Oersted discover about the relationship between electricity and magnetism?
2. List two uses of electromagnetism.
3. a) A uniform magnetic field of 0,35 T in the vertical direction exists. A piece of cardboard, of surface area 0,35 m^2 is placed flat on a horizontal surface inside the field. What is the magnetic flux through the cardboard?
   b) The one edge is then lifted so that the cardboard is now inclined at 17° to the positive x-direction. What is the magnetic flux through the cardboard? What will the induced emf be if the field drops to zero in the space of 3 s? Why?
4. A uniform magnetic field of 5 T in the vertical direction exists. What is the magnetic flux through a horizontal surface of area 0,68 m^2? What is the flux if the magnetic field changes to being in the positive x-direction?
5. A uniform magnetic field of 5 T in the vertical direction exists. What is the magnetic flux through a horizontal circle of radius 0,68 m?

6. Consider a square coil of 3 turns with a side length of 1,56 m. The coil is subjected to a varying magnetic field that changes uniformly from 4,38 T to 0,35 T in an interval of 3 minutes. The axis of the solenoid makes an angle of 197° to the magnetic field. Find the induced emf.

7. Consider a solenoid coil of 13 turns with radius $6,8 \times 10^{-2}$ m. The solenoid is subjected to a varying magnetic field that changes uniformly from -5 T to 1,8 T in an interval of 18 s. The axis of the solenoid makes an angle of 88° to the magnetic field.

   a) Find the induced emf.

   b) If the angle is changed to 39°, what would the radius need to be for the emf to remain the same?

8. Consider a solenoid with 5 turns and a radius of $4,3 \times 10^{-1}$ mm. The axis of the solenoid makes an angle of 11° to the magnetic field.

   Find the change in flux if the emf is 0,12 V over a period of 0,5 s.

9. Consider a rectangular coil of area 1,73 m^2. The coil is subjected to a varying magnetic field that changes uniformly from 2 T to 10 T in an interval of 3 ms. The axis of the solenoid makes an angle of 55° to the magnetic field. Find the induced emf.

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7. 242G 8. 242H 9. 242J

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CHAPTER 11

Electric circuits

11.1 Introduction ESBQ5

The study of electrical circuits is essential to understand the technology that uses electricity in the real-world. We depend on electricity and electrical appliances to make many things possible in our daily lives. This becomes very clear when there is a power failure and we can't use the kettle to boil water for tea or coffee, can't use the stove or oven to cook dinner, can't charge our cellphone batteries, watch TV, or use electric lights.

Key Mathematics Concepts

- Units and unit conversions — Physical Sciences, Grade 10, Science skills
- Equations — Mathematics, Grade 10, Equations and inequalities
- Graphs — Mathematics, Grade 10, Functions and graphs

11.2 Ohm's Law ESBQ6

Three quantities which are fundamental to electric circuits are **current, voltage (potential difference)** and **resistance**. To recap:

1. Electrical current, I, is defined as the rate of flow of charge through a circuit.
2. Potential difference or voltage, V, is the amount of energy per unit charge needed to move that charge between two points in a circuit.
3. Resistance, R, is a measure of how 'hard' it is to push current through a circuit element.

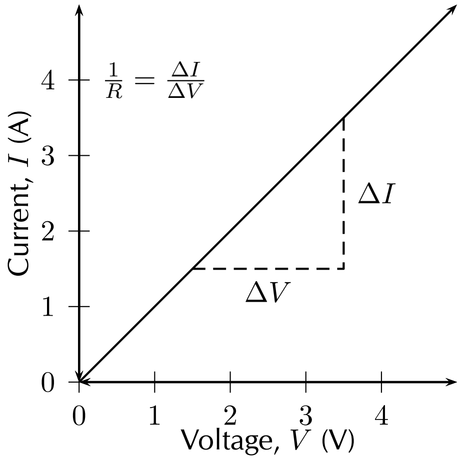
We will now look at how these three quantities are related to each other in electric circuits.

An important relationship between the current, voltage and resistance in a circuit was discovered by Georg Simon Ohm and it is called **Ohm's Law**.

DEFINITION: *Ohm's Law*

The amount of electric current through a metal conductor, at a constant temperature, in a circuit is proportional to the voltage across the conductor and can be described by
$I = \frac{V}{R}$
where I is the current through the conductor, V is the voltage across the conductor and R is the resistance of the conductor. In other words, at constant temperature, the resistance of the conductor is constant, independent of the voltage applied across it or current passed through it.

Ohm's Law tells us that if a conductor is at a constant temperature, the current flowing through the conductor is directly proportional to the voltage across it. This means that if we plot voltage on the x-axis of a graph and current on the y-axis of the graph, we will get a straight-line.



The gradient of the straight-line graph is related to the resistance of the conductor as

$$\frac{I}{V} = \frac{1}{R}.$$

This can be rearranged in terms of the constant resistance as:

$$R = \frac{V}{I}.$$

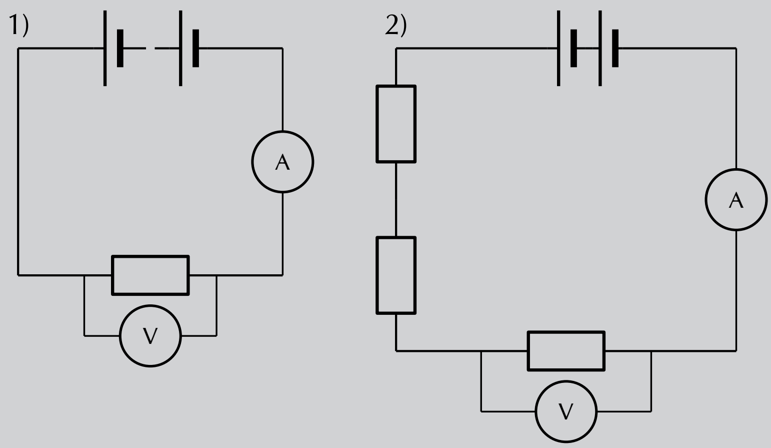
General experiment: Ohm's Law

Aim:

To determine the relationship between the current going through a resistor and the potential difference (voltage) across the same resistor.

Apparatus:

4 cells, 4 resistors, an ammeter, a voltmeter, connecting wires



Method:

This experiment has two parts. In the first part we will vary the applied voltage across the resistor and measure the resulting current through the circuit. In the second part we will vary the current in the circuit and measure the resulting voltage across the resistor. After obtaining both sets of measurements, we will examine the relationship between the current and the voltage across the resistor.

1. Varying the voltage:
   a) Set up the circuit according to circuit diagram 1), starting with just one cell.
   b) Draw the following table in your lab book.

Number of cells	Voltage, V (V)	Current, I (A)
1		
2		
3		
4		

   c) Get your teacher to check the circuit before turning the power on.
   d) Measure the voltage across the resistor using the voltmeter, and the current in the circuit using the ammeter.
   e) Add one more 1,5 V cell to the circuit and repeat your measurements.
   f) Repeat until you have four cells and you have completed your table.

2. Varying the current:
   a) Set up the circuit according to circuit diagram 2), starting with only 1 resistor in the circuit.
   b) Draw the following table in your lab book.

Voltage, V (V)	Current, I (A)

   c) Get your teacher to check your circuit before turning the power on.
   d) Measure the current and measure the voltage across the single resistor.
   e) Now add another resistor in series in the circuit and measure the current and the voltage across only the original resistor again. Continue adding resistors until you have four in series, but remember to only measure the voltage across the original resistor each time. Enter the values you measure into the table.

Analysis and results:

1. Using the data you recorded in the first table, draw a graph of current versus voltage. Since the voltage is the variable which we are directly varying, it is the independent variable and will be plotted on the x-axis. The current is the dependent variable and must be plotted on the y-axis.

2. Using the data you recorded in the second table, draw a graph of voltage vs. current. In this case the independent variable is the current which must be plotted on the x-axis, and the voltage is the dependent variable and must be plotted on the y-axis.

Conclusions:

1. Examine the graph you made from the first table. What happens to the current through the resistor when the voltage across it is increased? i.e. Does it increase or decrease?
2. Examine the graph you made from the second table. What happens to the voltage across the resistor when the current increases through the resistor? i.e. Does it increase or decrease?
3. Do your experimental results verify Ohm's Law? Explain.

Questions and discussion:

1. For each of your graphs, calculate the gradient and from this determine the resistance of the original resistor. Do you get the same value when you calculate it for each of your graphs?
2. How would you go about finding the resistance of an unknown resistor using only a power supply, a voltmeter and a known resistor R_0?

See simulation: 242K at www.everythingscience.co.za

Exercise 11 – 1: Ohm's Law

1. Use the data in the table below to answer the following questions.

Voltage, V (V)	Current, I (A)
3,0	0,4
6,0	0,8
9,0	1,2
12,0	1,6

   a) Plot a graph of voltage (on the x-axis) and current (on the y-axis).
   b) What type of graph do you obtain (straight-line, parabola, other curve)
   c) Calculate the gradient of the graph.
   d) Do your experimental results verify Ohm's Law? Explain.
   e) How would you go about finding the resistance of an unknown resistor using only a power supply, a voltmeter and a known resistor R_0?

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Ohmic and non-ohmic conductors ESBQ7

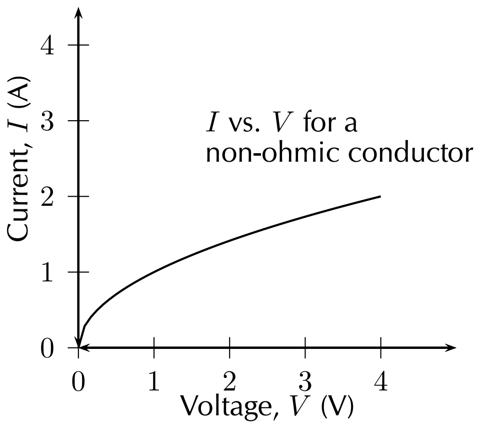
Conductors which obey Ohm's Law have a constant resistance when the voltage is varied across them or the current through them is increased. These conductors are called *ohmic* conductors. A graph of the current vs. the voltage across these conductors will be a straight-line. Some examples of ohmic conductors are circuit resistors and nichrome wire.

As you have seen, there is a mention of *constant temperature* when we talk about Ohm's Law. This is because the resistance of some conductors changes as their temperature changes. These types of conductors are called *non-ohmic* conductors, because they do not obey Ohm's Law. A light bulb is a common example of a non-ohmic conductor. Other examples of non-ohmic conductors are diodes and transistors.

In a light bulb, the resistance of the filament wire will increase dramatically as it warms from room temperature to operating temperature. If we increase the supply voltage in a real lamp circuit, the resulting increase in current causes the filament to increase in temperature, which increases its resistance. This effectively limits the increase in current. In this case, voltage and current do not obey Ohm's Law.

The phenomenon of resistance changing with variations in temperature is one shared by almost all metals, of which most wires are made. For most applications, these changes in resistance are small enough to be ignored. In the application of metal lamp filaments, which increase a lot in temperature (up to about 1000 °C, and starting from room temperature) the change is quite large.

In general, for non-ohmic conductors, a graph of voltage against current will not be a straight-line, indicating that the resistance is not constant over all values of voltage and current.



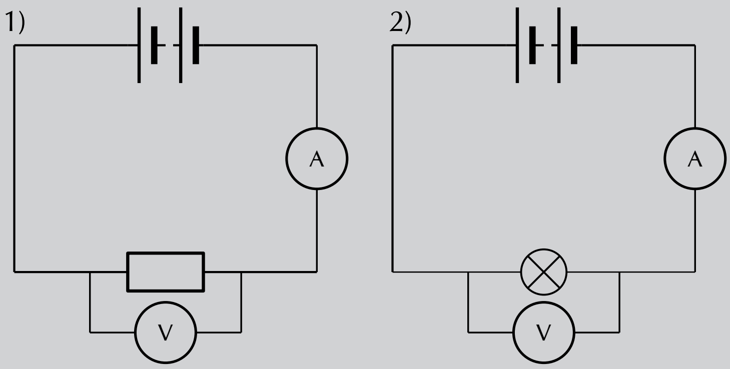
Informal experiment:Ohmic and non-ohmic conductors

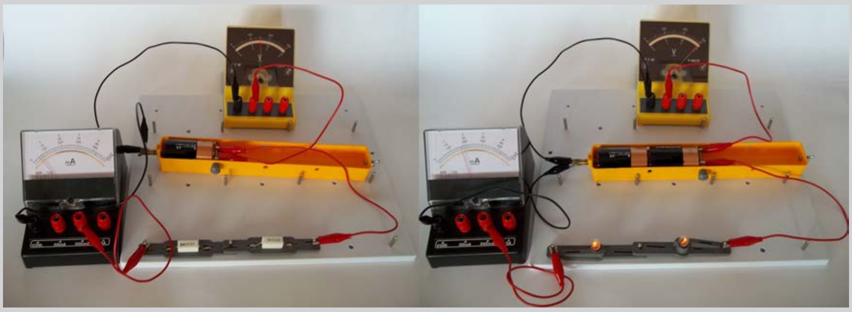
Aim:

To determine whether two circuit elements (a resistor and a lightbulb) obey Ohm's Law

Apparatus:

4 cells, a resistor, a lightbulb, connecting wires, a voltmeter, an ammeter





Method:

The two circuits shown in the diagrams above are the same, except in the first there is a resistor and in the second there is a lightbulb. Set up both the circuits above, starting with 1 cell. For each circuit:

1. Measure the voltage across the circuit element (either the resistor or lightbulb) using the voltmeter.
2. Measure the current in the circuit using the ammeter.
3. Add another cell and repeat your measurements until you have 4 cells in your circuit.

Results:

Draw two tables which look like the following in your book. You should have one table for the first circuit measurements with the resistor and another table for the second circuit measurements with the lightbulb.

Number of cells	Voltage, V (V)	Current, I (A)
1		
2		
3		
4		

Analysis:

Using the data in your tables, draw two graphs of I (y-axis) vs. V (x-axis), one for the resistor and one for the lightbulb.

Questions and Discussion:

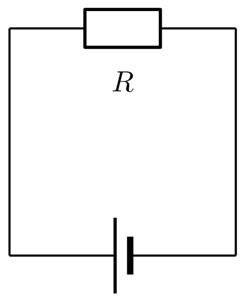
Examine your graphs closely and answer the following questions:

1. What should the graph of I vs. V look like for a conductor which obeys Ohm's Law?
2. Do either or both your graphs look like this?
3. What can you conclude about whether or not the resistor and/or the lightbulb obey Ohm's Law?
4. Is the lightbulb an ohmic or non-ohmic conductor?

Using Ohm's Law ESBQ8

We are now ready to see how Ohm's Law is used to analyse circuits.

Consider a circuit with a cell and an ohmic resistor, R. If the resistor has a resistance of 5 Ω and voltage across the resistor is 5 V, then we can use Ohm's Law to calculate the current flowing through the resistor. Our first task is to draw the circuit diagram. When solving any problem with electric circuits it is very important to make a diagram of the circuit before doing any calculations. The circuit diagram for this problem is shown alongside.



The equation for Ohm's Law is:

$$R = \frac{V}{I}$$

which can be rearranged to:

$$I = \frac{V}{R}$$

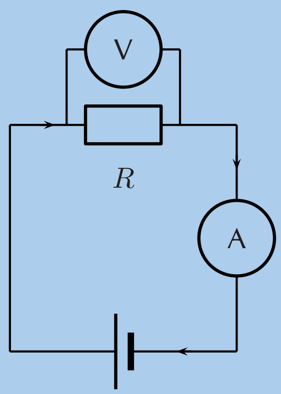
The current flowing through the resistor is:

$$
\begin{aligned}
I &= \frac{V}{R} \\
&= \frac{5\text{ V}}{5\Omega} \\
&= 1\text{ A}
\end{aligned}
$$

Worked example 1: Ohm's Law

QUESTION

Study the circuit diagram below:



The resistance of the resistor is 10 Ω and the current going through the resistor is 4 A. What is the potential difference (voltage) across the resistor?

SOLUTION

Step 1: Determine how to approach the problem

We are given the resistance of the resistor and the current passing through it and are asked to calculate the voltage across it. We can apply Ohm's Law to this problem using:

$$R = \frac{V}{I}.$$

Step 2: Solve the problem

Rearrange the equation above and substitute the known values for R and I to solve for V.

$$
\begin{aligned}
R &= \frac{V}{I} \\
R \times I &= \frac{V}{I} \times I \\
V &= I \times R \\
&= 10 \times 4 \\
&= 40\text{ V}
\end{aligned}
$$

Step 3: Write the final answer

The voltage across the resistor is 40 V.

Exercise 11 – 2: Ohm's Law

1. Calculate the resistance of a resistor that has a potential difference of 8 V across it when a current of 2 A flows through it. Draw the circuit diagram before doing the calculation.
2. What current will flow through a resistor of 6 Ω when there is a potential difference of 18 V across its ends? Draw the circuit diagram before doing the calculation.
3. What is the voltage across a 10 Ω resistor when a current of 1,5 A flows though it? Draw the circuit diagram before doing the calculation.

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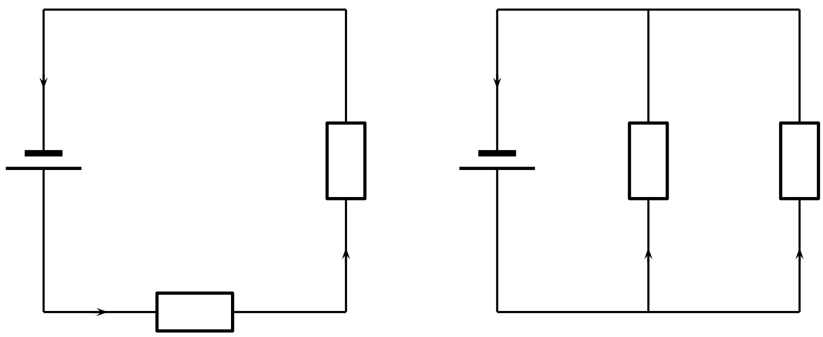
Recap of resistors in series and parallel ESBQ9

In Grade 10, you learnt about resistors and were introduced to circuits where resistors were connected in series and in parallel. In a series circuit there is one path along which current flows. In a parallel circuit there are multiple paths along which current flows.



series circuit
one current path

parallel circuit
multiple current paths

When there is more than one resistor in a circuit, we are usually able to calculate the total combined resistance of all the resistors. This is known as the *equivalent resistance*.

Equivalent series resistance

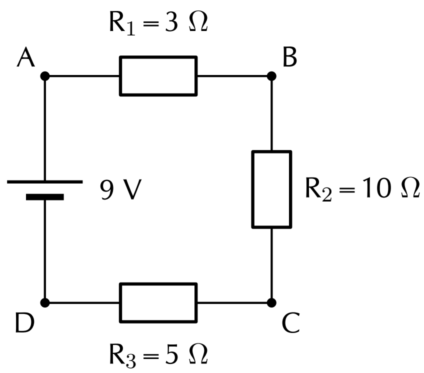
In a circuit where the resistors are connected in series, the equivalent resistance is just the sum of the resistances of all the resistors.

DEFINITION: *Equivalent resistance in a series circuit,*

For n resistors in series the equivalent resistance is:

$$R_s = R_1 + R_2 + R_3 + \ldots + R_n$$

Let us apply this to the following circuit.



The resistors are in series, therefore:

$$\begin{aligned} R_s &= R_1 + R_2 + R_3 \\ &= 3\ \Omega + 10\ \Omega + 5\ \Omega \\ &= 18\ \Omega \end{aligned}$$

See simulation: 242R at www.everythingscience.co.za

See video: 242S at www.everythingscience.co.za

Equivalent parallel resistance

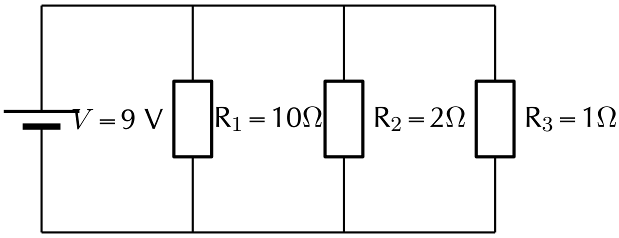
In a circuit where the resistors are connected in parallel, the equivalent resistance is given by the following definition.

DEFINITION: *Equivalent resistance in a parallel circuit*

For n resistors in parallel, the equivalent resistance is:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n}$$

Let us apply this formula to the following circuit.



What is the total (equivalent) resistance in the circuit?

$$\begin{aligned}
\frac{1}{R_p} &= \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)\\
&= \left(\frac{1}{10\ \Omega} + \frac{1}{2\ \Omega} + \frac{1}{1\ \Omega}\right)\\
&= \left(\frac{1\ \Omega + 5\ \Omega + 10\ \Omega}{10\ \Omega}\right)\\
&= \left(\frac{16\ \Omega}{10\ \Omega}\right)\\
R_p &= 0{,}625\ \Omega
\end{aligned}$$

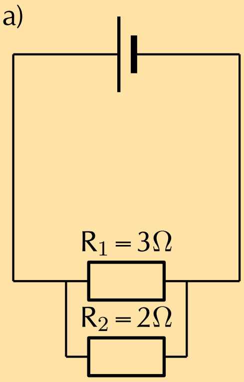
See video: 242T at www.everythingscience.co.za

See video: 242V at www.everythingscience.co.za

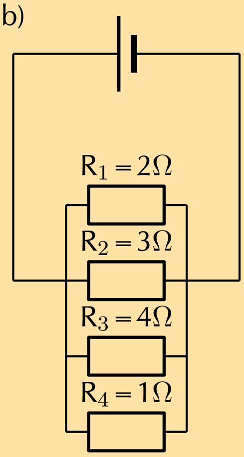
Exercise 11 – 3: Series and parallel resistance

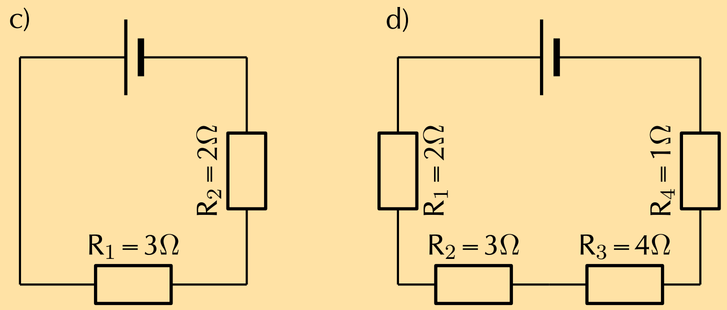
1. Two 10 kΩ resistors are connected in series. Calculate the equivalent resistance.
2. Two resistors are connected in series. The equivalent resistance is 100 Ω. If one resistor is 10 Ω, calculate the value of the second resistor.
3. Two 10 kΩ resistors are connected in parallel. Calculate the equivalent resistance.
4. Two resistors are connected in parallel. The equivalent resistance is 3,75 Ω. If one resistor has a resistance of 10 Ω, what is the resistance of the second resistor?
5. Calculate the equivalent resistance in each of the following circuits:

a)



b)





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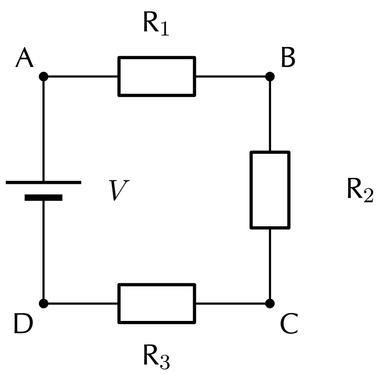
Use of Ohm's Law in series and parallel circuits ESBQB

Using the definitions for equivalent resistance for resistors in series or in parallel, we can analyse some circuits with these setups.

Series circuits

Consider a circuit consisting of three resistors and a single cell connected in series.



The first principle to understand about series circuits is that the amount of current is the same through any component in the circuit. This is because there is only one path for electrons to flow in a series circuit. From the way that the battery is connected, we can tell in which direction the current will flow. We know that current flows from positive to negative by convention. Conventional current in this circuit will flow in a clockwise direction, from point A to B to C to D and back to A.

We know that in a series circuit the current has to be the same in all components. So we can write:

$$I = I_1 = I_2 = I_3.$$

We also know that total voltage of the circuit has to be equal to the sum of the voltages over all three resistors. So we can write:

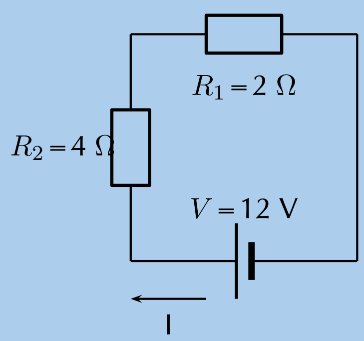
$$V = V_1 + V_2 + V_3$$

Using this information and what we know about calculating the equivalent resistance of resistors in series, we can approach some circuit problems.

Worked example 2: Ohm's Law, series circuit

QUESTION

Calculate the current (I) in this circuit if the resistors are both ohmic in nature.



SOLUTION

Step 1: Determine what is required

We are required to calculate the current flowing in the circuit.

Step 2: Determine how to approach the problem

Since the resistors are ohmic in nature, we can use Ohm's Law. There are however two resistors in the circuit and we need to find the total resistance.

Step 3: Find total resistance in circuit

Since the resistors are connected in series, the total (equivalent) resistance R is:

$$R = R_1 + R_2$$

Therefore,

$$\begin{aligned} R &= 2 + 4 \\ &= 6\ \Omega \end{aligned}$$

Step 4: Apply Ohm's Law

$$R = \frac{V}{I}$$
$$R \times \frac{I}{R} = \frac{V}{I} \times \frac{I}{R}$$
$$I = \frac{V}{R}$$
$$= \frac{12}{6}$$
$$= 2 \text{ A}$$

Step 5: Write the final answer

A current of 2 A is flowing in the circuit.

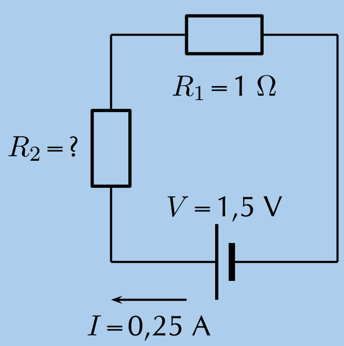
Worked example 3: Ohm's Law, series circuit

QUESTION

Two ohmic resistors (R_1 and R_2) are connected in series with a cell. Find the resistance of R_2, given that the current flowing through R_1 and R_2 is 0,25 A and that the voltage across the cell is 1,5 V. $R_1 = 1\ \Omega$.

SOLUTION

Step 1: Draw the circuit and fill in all known values.



Step 2: Determine how to approach the problem.

We can use Ohm's Law to find the total resistance R in the circuit, and then calculate the unknown resistance using:

$$R = R_1 + R_2$$

because it is in a series circuit.

Step 3: Find the total resistance

$$R = \frac{V}{I}$$
$$= \frac{1,5}{0,25}$$
$$= 6\ \Omega$$

Step 4: Find the unknown resistance

We know that:

$$R = 6\ \Omega$$

and that

$$R_1 = 1\ \Omega$$

Since

$$R = R_1 + R_2$$
$$R_2 = R - R_1$$

Therefore,

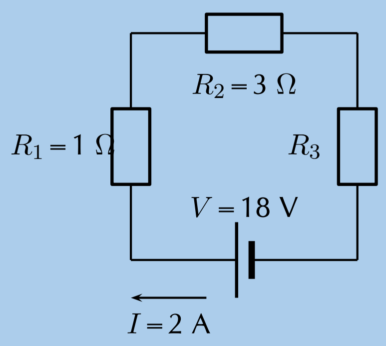
$$R_1 = 5\ \Omega$$

Worked example 4: Ohm's Law, series circuit

QUESTION

For the following circuit, calculate:

1. the voltage drops V_1, V_2 and V_3 across the resistors R_1, R_2, and R_3
2. the resistance of R_3.



SOLUTION

Step 1: Determine how to approach the problem

We are given the voltage across the cell and the current in the circuit, as well as the resistances of two of the three resistors. We can use Ohm's Law to calculate the voltage drop across the known resistors. Since the resistors are in a series circuit the voltage is $V = V_1 + V_2 + V_3$ and we can calculate V_3. Now we can use this information to find the voltage across the unknown resistor R_3.

Step 2: Calculate voltage drop across R_1

Using Ohm's Law:

$$\begin{aligned} R_1 &= \frac{V_1}{I} \\ I \cdot R_1 &= I \cdot \frac{V_1}{I} \\ V_1 &= I \cdot R_1 \\ &= 2 \cdot 1 \\ V_1 &= 2 \text{ V} \end{aligned}$$

Step 3: Calculate voltage drop across R_2

Again using Ohm's Law:

$$\begin{aligned} R_2 &= \frac{V_2}{I} \\ I \cdot R_2 &= I \cdot \frac{V_2}{I} \\ V_2 &= I \cdot R_2 \\ &= 2 \cdot 3 \\ V_2 &= 6 \text{ V} \end{aligned}$$

Step 4: Calculate voltage drop across R_3

Since the voltage drop across all the resistors combined must be the same as the voltage drop across the cell in a series circuit, we can find V_3 using:

$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ V_3 &= V - V_1 - V_2 \\ &= 18 - 2 - 6 \\ V_3 &= 10 \text{ V} \end{aligned}$$

Step 5: Find the resistance of R_3

We know the voltage across R_3 and the current through it, so we can use Ohm's Law to calculate the value for the resistance:

$$\begin{aligned} R_3 &= \frac{V_3}{I} \\ &= \frac{10}{2} \\ R_3 &= 5\Omega \end{aligned}$$

Step 6: Write the final answer

$V_1 = 2$ V

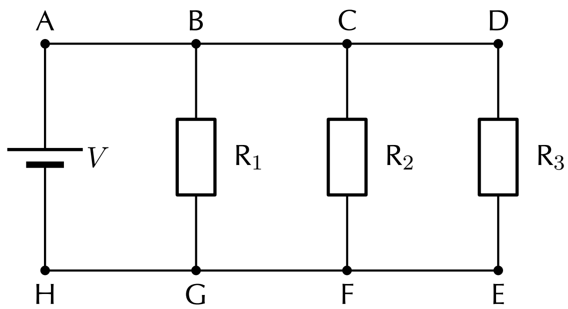
$V_2 = 6$ V

$V_3 = 10$ V

$R_1 = 5\Omega$

Parallel circuits

Consider a circuit consisting of a single cell and three resistors that are connected in parallel.



The first principle to understand about parallel circuits is that the voltage is equal across all components in the circuit. This is because there are only two sets of electrically common points in a parallel circuit, and voltage measured between sets of common points must always be the same at any given time. So, for the circuit shown, the following is true:

$$V = V_1 = V_2 = V_3.$$

The second principle for a parallel circuit is that all the currents through each resistor must add up to the total current in the circuit:

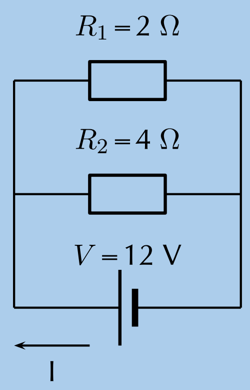
$$I = I_1 + I_2 + I_3.$$

Using these principles and our knowledge of how to calculate the equivalent resistance of parallel resistors, we can now approach some circuit problems involving parallel resistors.

Worked example 5: Ohm's Law, parallel circuit

QUESTION

Calculate the current (I) in this circuit if the resistors are both ohmic in nature.



SOLUTION

Step 1: Determine what is required

We are required to calculate the current flowing in the circuit.

Step 2: Determine how to approach the problem

Since the resistors are ohmic in nature, we can use Ohm's Law. There are however two resistors in the circuit and we need to find the total resistance.

Step 3: Find the equivalent resistance in circuit

Since the resistors are connected in parallel, the total (equivalent) resistance R is:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

$$\begin{aligned}\frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2}\\ &= \frac{1}{2} + \frac{1}{4}\\ &= \frac{2+1}{4}\\ &= \frac{3}{4}\\ \text{Therefore, } R &= 1{,}33\Omega\end{aligned}$$

Step 4: Apply Ohm's Law

$$
\begin{aligned}
R &= \frac{V}{I} \\
R \cdot \frac{I}{R} &= \frac{V}{I} \cdot \frac{I}{R} \\
I &= \frac{V}{R} \\
I &= V \cdot \frac{1}{R} \\
&= (12)\left(\frac{3}{4}\right) \\
&= 9\ \text{A}
\end{aligned}
$$

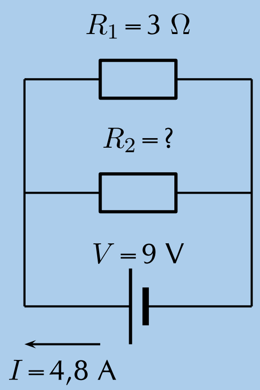
Step 5: Write the final answer

The current flowing in the circuit is 9 A.

Worked example 6: Ohm's Law, parallel circuit

QUESTION

Two ohmic resistors (R_1 and R_2) are connected in parallel with a cell. Find the resistance of R_2, given that the current flowing through the cell is 4,8 A and that the voltage across the cell is 9 V.



SOLUTION

Step 1: Determine what is required

We need to calculate the resistance R_2.

Step 2: Determine how to approach the problem

Since the resistors are ohmic and we are given the voltage across the cell and the current through the cell, we can use Ohm's Law to find the equivalent resistance in the circuit.

$$
\begin{aligned}
R &= \frac{V}{I} \\
&= \frac{9}{4{,}8} \\
&= 1{,}875\ \Omega
\end{aligned}
$$

Step 3: Calculate the value for R_2

Since we know the equivalent resistance and the resistance of R_1, we can use the formula for resistors in parallel to find the resistance of R_2.

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Rearranging to solve for R_2:

$$
\begin{aligned}
\frac{1}{R_2} &= \frac{1}{R} - \frac{1}{R_1} \\
&= \frac{1}{1{,}875} - \frac{1}{3} \\
&= 0{,}2 \\
R_2 &= \frac{1}{0{,}2} \\
&= 5\ \Omega
\end{aligned}
$$

Step 4: Write the final answer

The resistance R_2 is 5 Ω

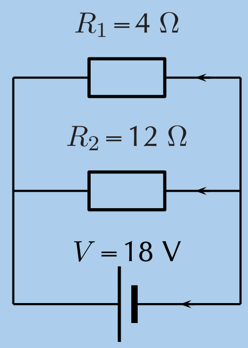
Worked example 7: Ohm's Law, parallel circuit

QUESTION

An 18 volt cell is connected to two parallel resistors of 4 Ω and 12 Ω respectively. Calculate the current through the cell and through each of the resistors.

SOLUTION

Step 1: First draw the circuit before doing any calculations



Step 2: Determine how to approach the problem

We need to determine the current through the cell and each of the parallel resistors. We have been given the potential difference across the cell and the resistances of the resistors, so we can use Ohm's Law to calculate the current.

Step 3: Calculate the current through the cell

To calculate the current through the cell we first need to determine the equivalent resistance of the rest of the circuit. The resistors are in parallel and therefore:

$$\begin{aligned}\frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2}\\ &= \frac{1}{4} + \frac{1}{12}\\ &= \frac{3+1}{12}\\ &= \frac{4}{12}\\ R &= \frac{12}{4} = 3\ \Omega\end{aligned}$$

Now using Ohm's Law to find the current through the cell:

$$\begin{aligned}R &= \frac{V}{I}\\ I &= \frac{V}{R}\\ &= \frac{18}{3}\\ I &= 6\text{ A}\end{aligned}$$

Step 4: Now determine the current through one of the parallel resistors

We know that for a purely parallel circuit, the voltage across the cell is the same as the voltage across each of the parallel resistors. For this circuit:

$$V = V_1 = V_2 = 18\text{ V}$$

Let's start with calculating the current through R_1 using Ohm's Law:

$$\begin{aligned} R_1 &= \frac{V_1}{I_1} \\ I_1 &= \frac{V_1}{R_1} \\ &= \frac{18}{4} \\ I_1 &= 4{,}5 \text{ A} \end{aligned}$$

Step 5: Calculate the current through the other parallel resistor

We can use Ohm's Law again to find the current in R_2:

$$\begin{aligned} R_2 &= \frac{V_2}{I_2} \\ I_2 &= \frac{V_2}{R_2} \\ &= \frac{18}{12} \\ I_2 &= 1{,}5 \text{ A} \end{aligned}$$

An alternative method of calculating I_2 would have been to use the fact that the currents through each of the parallel resistors must add up to the total current through the cell:

$$\begin{aligned} I &= I_1 + I_2 \\ I_2 &= I - I_1 \\ &= 6 - 4.5 \\ I_2 &= 1{,}5 \text{ A} \end{aligned}$$

Step 6: Write the final answer

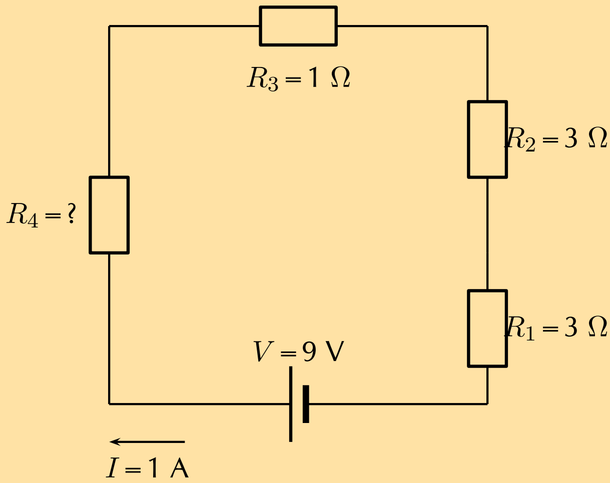
The current through the cell is 6 A.

The current through the 4 Ω resistor is 4,5 A.

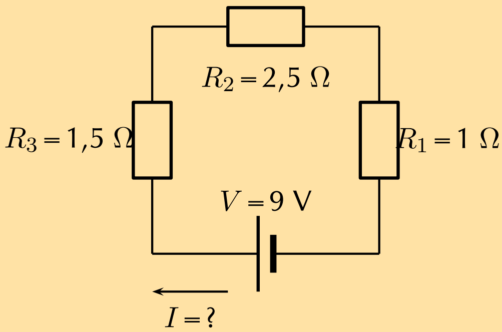
The current through the 12 Ω resistor is 1,5 A.

Exercise 11 – 4: Ohm's Law in series and parallel circuits

1. Calculate the value of the unknown resistor in the circuit:

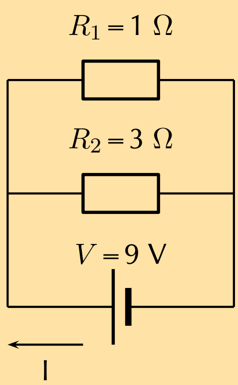


2. Calculate the value of the current in the following circuit:

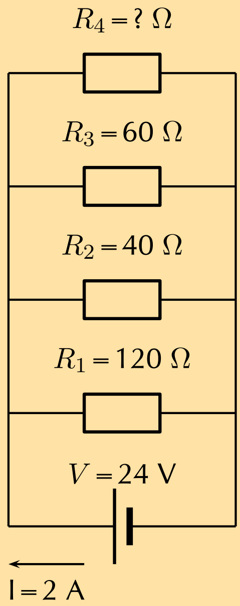


3. Three resistors with resistance 1 Ω, 5 Ω and 10 Ω respectively, are connected in series with a 12 V battery. Calculate the value of the current in the circuit.

4. Calculate the current through the cell if the resistors are both ohmic in nature.



5. Calculate the value of the unknown resistor R_4 in the circuit:



6. Three resistors with resistance 1 Ω, 5 Ω and 10 Ω respectively, are connected in parallel with a 20 V battery. All the resistors are ohmic in nature. Calculate:

   a) the value of the current through the battery

   b) the value of the current in each of the three resistors.

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Series and parallel networks of resistors ESBQC

Now that you know how to handle simple series and parallel circuits, you are ready to tackle circuits which combine these two setups such as the following circuit:

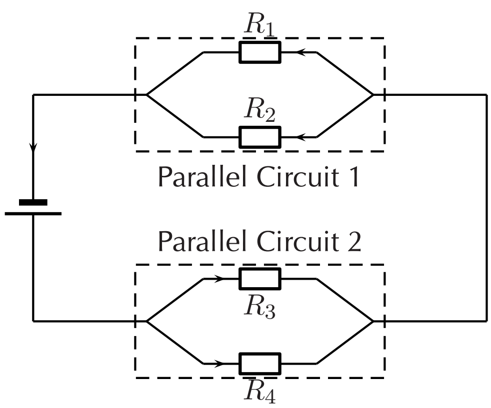
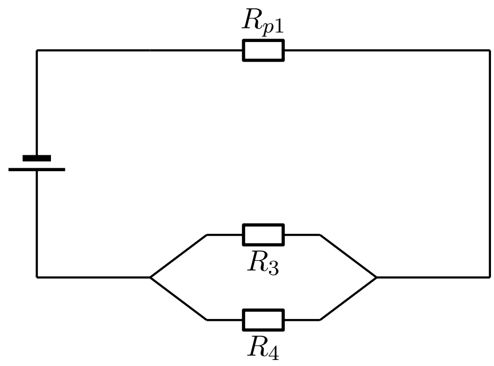


Figure 11.1: An example of a series-parallel network. The dashed boxes indicate parallel sections of the circuit.

It is relatively easy to work out these kind of circuits because you use everything you have already learnt about series and parallel circuits. The only difference is that you do it in stages. In figure 11.1, the circuit consists of 2 parallel portions that are then in series with a cell. To work out the equivalent resistance for the circuit, you start by calculating the total resistance of each of the parallel portions and then add up these resistances in series. If all the resistors in figure 11.1 had resistances of 10 Ω, we can calculate the equivalent resistance of the entire circuit.

We start by calculating the total resistance of *Parallel Circuit 1*.



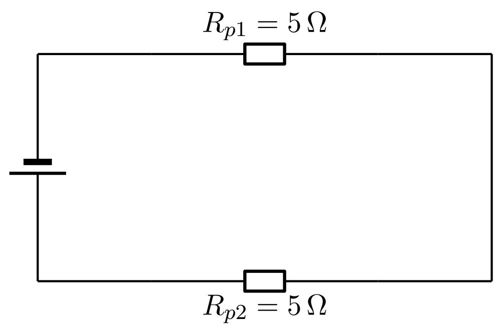
The value of R_{p1} is:

$$\begin{aligned}\frac{1}{R_{p1}} &= \frac{1}{R_1} + \frac{1}{R_2}\\ R_{p1} &= \left(\frac{1}{10} + \frac{1}{10}\right)^{-1}\\ &= \left(\frac{1+1}{10}\right)^{-1}\\ &= \left(\frac{2}{10}\right)^{-1}\\ &= 5\,\Omega\end{aligned}$$

We can similarly calculate the total resistance of *Parallel Circuit 2*:

$$\begin{aligned}\frac{1}{R_{p2}} &= \frac{1}{R_3} + \frac{1}{R_4}\\ R_{p2} &= \left(\frac{1}{10} + \frac{1}{10}\right)^{-1}\\ &= \left(\frac{1+1}{10}\right)^{-1}\\ &= \left(\frac{2}{10}\right)^{-1}\\ &= 5\,\Omega\end{aligned}$$

You can now treat the circuit like a simple series circuit as follows:



Therefore the equivalent resistance is:

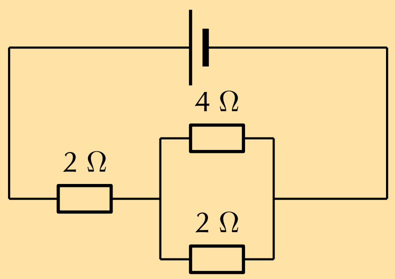
$$
\begin{aligned}
R &= R_{p1} + R_{p2} \\
&= 5 + 5 \\
&= 10\,\Omega
\end{aligned}
$$

The equivalent resistance of the circuit in figure 11.1 is 10 Ω.

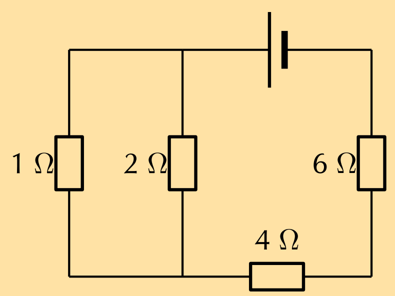
Exercise 11 – 5: Series and parallel networks

1. Determine the equivalent resistance of the following circuits:

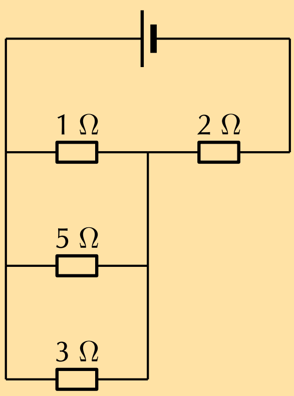
a)



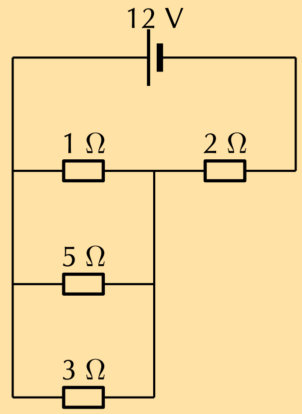
b)



c)



2. Examine the circuit below:

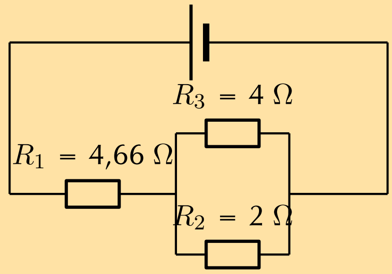


If the potential difference across the cell is 12 V, calculate:

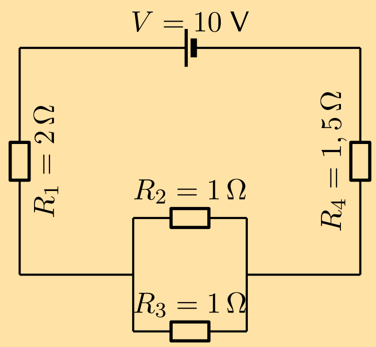
a) the current I through the cell.

b) the current through the 5 Ω resistor.

3. If current flowing through the cell is 2 A, and all the resistors are ohmic, calculate the voltage across the cell and each of the resistors, R_1, R_2, and R_3 respectively.



4. For the following circuit, calculate:



a) the current through the cell

b) the voltage drop across R_4

c) the current through R_2

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11.3 Power and energy ESBQD

Electrical power ESBQF

A source of energy is required to drive current round a complete circuit. This is provided by batteries in the circuits you have been looking at. The batteries convert chemical potential energy into electrical energy. The energy is used to do work on the electrons in the circuit.

Power is a measure of how rapidly *work* is done. Power is the **rate** at which the work is done, work done per unit time. Work is measured in joules (J) and time in seconds (s) so power will be $\frac{\text{J}}{\text{s}}$ which we call a watt (W).

In electric circuits, power is a function of both voltage and current and we talk about the power *dissipated* in a circuit element:

DEFINITION: *Electrical Power*

Electrical power is the rate at which electrical energy is converted in an electric circuit. It calculated as:

$$P = I \cdot V$$

Power (P) is exactly equal to current (I) multiplied by voltage (V), there is no extra constant of proportionality. The unit of measurement for power is the watt (abbreviated W).

FACT
It was James Prescott Joule, not Georg Simon Ohm, who first discovered the mathematical relationship between power dissipation and current through a resistance. This discovery, published in 1841, followed the form of the equation: $P = I^2R$ and is properly known as Joule's Law. However, these power equations are so commonly associated with the Ohm's Law equations relating voltage, current, and resistance that they are frequently credited to Ohm.

Equivalent forms

We can use Ohm's Law to show that $P = VI$ is equivalent to $P = I^2R$ and $P = \frac{V^2}{R}$.

Using $V = I \cdot R$ allows us to show:

$$\begin{aligned} P &= V \cdot I \\ &= (I \cdot R) \cdot I \text{ Ohm's Law} \\ &= I^2 R \end{aligned}$$

Using $I = \frac{V}{R}$ allows us to show:

$$\begin{aligned} P &= V \cdot I \\ &= V \cdot \frac{V}{R} \text{ Ohm's Law} \\ &= \frac{V^2}{R} \end{aligned}$$

Worked example 8: Electrical power

QUESTION

Given a circuit component that has a voltage of 5 V and a resistance of 2 Ω what is the power dissipated?

SOLUTION

Step 1: Write down what you are given and what you need to find

$$V = 5 \text{ V}$$
$$R = 2\ \Omega$$
$$P = ?$$

Step 2: Write down an equation for power

The equation for power is:

$$P = V^2 R$$

Step 3: Solve the problem

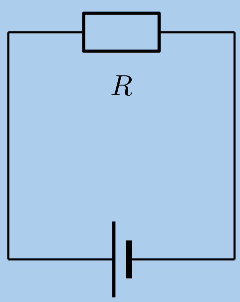
$$\begin{aligned} P &= \frac{V^2}{R} \\ &= \frac{(5)^2}{(2)} \\ &= 12{,}5 \text{ W} \end{aligned}$$

The power is 12,5 W.

Worked example 9: Electrical power

QUESTION

Study the circuit diagram below:



The resistance of the resistor is 15 Ω and the current going through the resistor is 4 A. What is the power for the resistor?

SOLUTION

Step 1: Determine how to approach the problem

We are given the resistance of the resistor and the current passing through it and are asked to calculate the power. We can have verified that:

$$P = I^2 R$$

Step 2: Solve the problem

We can simply substitute the known values for R and I to solve for P.

$$\begin{aligned} P &= I^2 R \\ &= (4)^2 \times 15 \\ &= 240 \text{ W} \end{aligned}$$

Step 3: Write the final answer

The power for the resistor is 240 W.

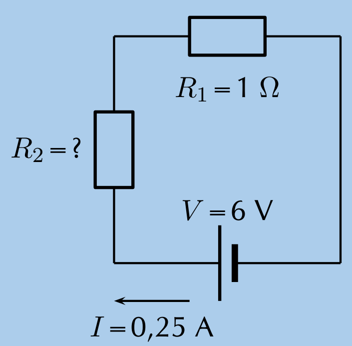
Worked example 10: Power in series circuit

QUESTION

Two ohmic resistors (R_1 and R_2) are connected in series with a cell. Find the resistance and power of R_2, given that the current flowing through R_1 and R_2 is 0,25 A and that the voltage across the cell is 6 V. $R_1 = 1\ \Omega$.

SOLUTION

Step 1: Draw the circuit and fill in all known values.



Step 2: Determine how to approach the problem.

TIP
Notice that we use the same circuits in examples as we extend our knowledge of electric circuits. This is to emphasise that you can always combine all of the principles you have learnt when dealing with any circuit.

We can use Ohm's Law to find the total resistance R in the circuit, and then calculate the unknown resistance using:

$$R = R_1 + R_2$$

because it is in a series circuit.

Step 3: Find the total resistance

$$\begin{aligned} V &= R \cdot I \\ R &= \frac{V}{I} \\ &= \frac{6}{0{,}25} \\ &= 24\ \Omega \end{aligned}$$

Step 4: Find the unknown resistance

We know that:

$$R = 24\ \Omega$$

and that

$$R_1 = 1\ \Omega$$

Since

$$\begin{aligned} R &= R_1 + R_2 \\ R_2 &= R - R_1 \end{aligned}$$

Therefore,

$$R = 23\ \Omega$$

Step 5: Solve the problem

Now that the resistance is known and the current, we can determine the power:

$$\begin{aligned} P &= I^2 R \\ &= (0{,}25)^2(23) \\ &= 1{,}44\ \text{W} \end{aligned}$$

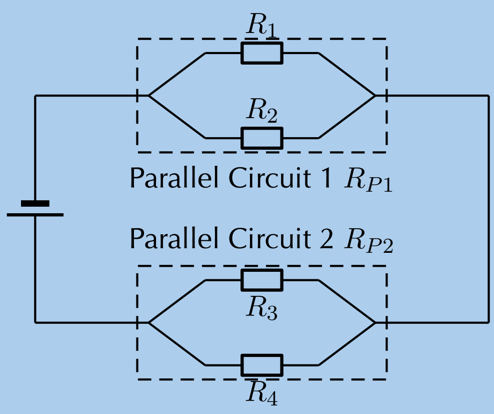
Step 6: Write the final answer

The power for the resistor R_2 is 1,44 W.

Worked example 11: Power in series and parallel networks of resistors

QUESTION

Given the following circuit:



The current leaving the battery is 1,07 A, the total power dissipated in the circuit is 6,42 W, the ratio of the total resistances of the two parallel networks $R_{P1} : R_{P2}$ is 1:2, the ratio $R_1 : R_2$ is 3:5 and $R_3 = 7\ \Omega$.

Determine the:

1. voltage of the battery,
2. the power dissipated in R_{P1} and R_{P2}, and
3. the value of each resistor and the power dissipated in each of them.

SOLUTION

Step 1: What is required

In this question you are given various pieces of information and asked to determine the power dissipated in each resistor and each combination of resistors. Notice that the information given is mostly for the overall circuit. This is a clue that you should start with the overall circuit and work downwards to more specific circuit elements.

Step 2: Calculating the voltage of the battery

Firstly we focus on the battery. We are given the power for the overall circuit as well as the current leaving the battery. We know that the voltage across the terminals of the battery is the voltage across the circuit as a whole.

We can use the relationship $P = VI$ for the entire circuit because the voltage is the same as the voltage across the terminals of the battery:

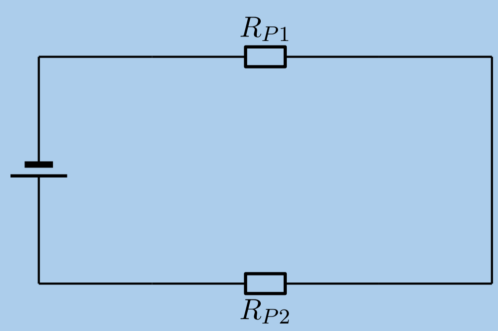
$$\begin{aligned} P &= VI \\ V &= \frac{P}{I} \\ &= \frac{6{,}42}{1{,}07} \\ &= 6{,}00 \text{ V} \end{aligned}$$

The voltage across the battery is 6,00 V.

Step 3: Power dissipated in R_{P1} and R_{P2}

Remember that we are working from the overall circuit details down towards those for individual elements, this is opposite to how you treated this circuit earlier.

We can treat the parallel networks like the equivalent resistors so the circuit we are currently dealing with looks like:



We know that the current through the two circuit elements will be the same because it is a series circuit and that the resistance for the total circuit must be: $R_T = R_{P1} + R_{P2}$. We can determine the total resistance from Ohm's Law for the circuit as a whole:

$$\begin{aligned} V_{battery} &= IR_T \\ R_T &= \frac{V_{battery}}{I} \\ &= \frac{6{,}00}{1{,}07} \\ &= 5{,}61\ \Omega \end{aligned}$$

We know that the ratio between $R_{P1} : R_{P2}$ is 1:2 which means that we know:

$$\begin{aligned} R_{P1} &= \frac{1}{2}R_{P2} \text{ and} \\ R_T &= R_{P1} + R_{P2} \\ &= \frac{1}{2}R_{P2} + R_{P2} \\ &= \frac{3}{2}R_{P2} \\ (5{,}61) &= \frac{3}{2}R_{P2} \\ R_{P2} &= \frac{2}{3}(5{,}61) \\ R_{P2} &= 3{,}74\ \Omega \end{aligned}$$

and therefore:

$$\begin{aligned} R_{P1} &= \frac{1}{2}R_{P2} \\ &= \frac{1}{2}(3.74) \\ &= 1{,}87\ \Omega \end{aligned}$$

Now that we know the total resistance of each of the parallel networks we can calculate the power dissipated in each:

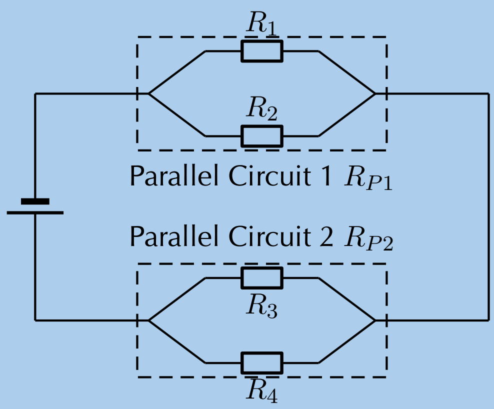
$$\begin{aligned} P_{P1} &= I^2 R_{P1} \\ &= (1{,}07)^2(1{,}87) \\ &= 2{,}14\ \text{W} \end{aligned}$$

and

$$\begin{aligned} P_{P2} &= I^2 R_{P2} \\ &= (1{,}07)^2(3{,}74) \\ &= 4{,}28\ \text{W} \end{aligned}$$

Step 4: Parallel network 1 calculations

Now we can begin to do the detailed calculation for the first set of parallel resistors.



We know that the ratio between $R_1 : R_2$ is 3:5 which means that we know $R_1 = \frac{3}{5}R_2$. We also know the total resistance for the two parallel resistors in this network is 1,87 Ω. We can use the relationship between the values of the two resistors as well as the formula for the total resistance ($\frac{1}{R_PT} = \frac{1}{R_1} + \frac{1}{R_2}$)to find the resistor values:

$$
\begin{aligned}
\frac{1}{R_{P1}} &= \frac{1}{R_1} + \frac{1}{R_2} \\
\frac{1}{R_{P1}} &= \frac{5}{3R_2} + \frac{1}{R_2} \\
\frac{1}{R_{P1}} &= \frac{1}{R_2}(\frac{5}{3} + 1) \\
\frac{1}{R_{P1}} &= \frac{1}{R_2}(\frac{5}{3} + \frac{3}{3}) \\
\frac{1}{R_{P1}} &= \frac{1}{R_2}\frac{8}{3} \\
R_2 &= R_{P1}\frac{8}{3} \\
&= (1{,}87)\frac{8}{3} \\
&= 4{,}99\ \Omega
\end{aligned}
$$

We can also calculate R_1:

$$
\begin{aligned}
R_1 &= \frac{3}{5}R_2 \\
&= \frac{3}{5}(4{,}99) \\
&= 2{,}99\ \Omega
\end{aligned}
$$

To determine the power we need the resistance which we have calculated and either the voltage or current. The two resistors are in parallel so the voltage across them is the same as well as the same as the voltage across the parallel network. We can use Ohm's Law to determine the voltage across the network of parallel resistors as we know the total resistance and we know the current:

$$
\begin{aligned}
V &= IR \\
&= (1{,}07)(1{,}87) \\
&= 2{,}00\ \text{V}
\end{aligned}
$$

We now have the information we need to determine the power through each resistor:

$$
\begin{aligned}
P_1 &= \frac{V^2}{R_1} \\
&= \frac{(2{,}00)^2}{2{,}99} \\
&= 1{,}34\ \text{W}
\end{aligned}
$$

$$
\begin{aligned}
P_2 &= \frac{V^2}{R_2} \\
&= \frac{(2{,}00)^2}{4{,}99} \\
&= 0{,}80\ \text{W}
\end{aligned}
$$

Step 5: Parallel network 2 calculations

Now we can begin to do the detailed calculation for the second set of parallel resistors.

We are given $R_3 = 7{,}00\ \Omega$ and we know R_{P2} so we can calculate R_4 from:

$$\frac{1}{R_{P2}} = \frac{1}{R_3} + \frac{1}{R_4}$$
$$\frac{1}{3{,}74} = \frac{1}{7{,}00} + \frac{1}{R_4}$$
$$R_4 = 8{,}03\ \Omega$$

We can calculate the voltage across the second parallel network by subtracting the voltage of the first parallel network from the battery voltage, $V_{P2} = 6{,}00 - 2{,}00 = 4{,}00$ V.

We can now determine the power dissipated in each resistor:

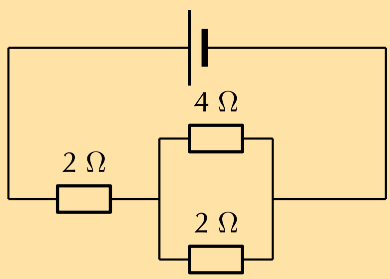
$$P_3 = \frac{V^2}{R_3} = \frac{(4{,}00)^2}{7{,}00} = 2{,}29\ \text{W}$$

$$P_4 = \frac{V^2}{R_2} = \frac{(4{,}00)^2}{8{,}03} = 1{,}99\ \text{W}$$

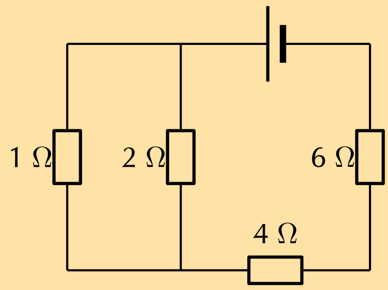
Exercise 11 – 6:

1. What is the power of a $1{,}00 \times 10^8$ V lightning bolt having a current of $2{,}00 \times 10^4$ A?
2. How many watts does a torch that has $6{,}00 \times 10^2$ C pass through it in 0,50 h use if its voltage is 3,00 V?
3. Find the power dissipated in each of these extension cords:
   a) an extension cord having a 0,06 Ω resistance and through which 5,00 A is flowing
   b) a cheaper cord utilising (using) thinner wire and with a resistance of 0,30 Ω, through which 5,00 A is flowing
4. Determine the power dissipated by each the resistors in the following circuits, if the batteries are 6 V:

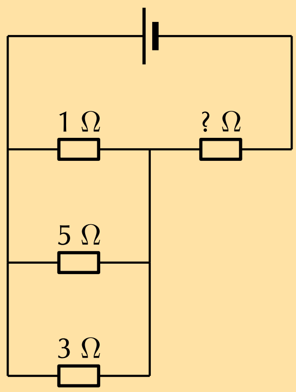
   a)



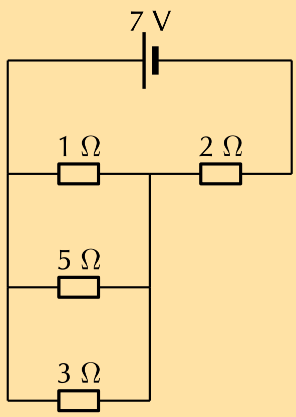
   b)



c) Also determine the value of the unknown resistor if the total power dissipated is 9,8 W



5. Examine the circuit below:



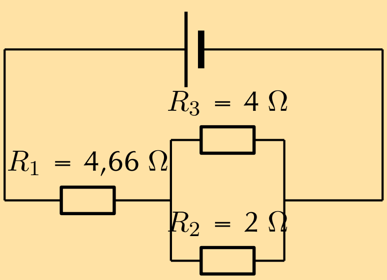
If the potential difference across the cell is 7 V, calculate:

a) the current I through the cell.

b) the current through the 5 Ω resistor

c) the power dissipated in the 5 Ω resistor

6. If current flowing through the cell is 2 A, and all the resistors are ohmic, calculate the power dissipated in each of the resistors, R_1, R_2, and R_3 respectively.



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1. 243H 2. 243J 3. 243K 4a. 243M 4b. 243N 4c. 243P
5. 243Q 6. 243R

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Electrical energy ESBQG

When power is dissipated in a device there is a transfer of energy from one kind to another. For example, a resistor may get very hot which indicates that the energy is being dissipated as heat. Power was the rate at which work was done, the rate at which energy is transferred. If we want to calculate the total amount of energy we need to multiply the rate of energy transfer by the time over which that energy transfer took place.

Electrical energy is simply power times time. Mathematically we write:

$$E = P \times t$$

Energy is measured in joules (J) and time in seconds (s).

Worked example 12: Electrical energy

QUESTION

A 30 W light bulb is left on for 8 hours overnight, how much energy was wasted?

SOLUTION

Step 1: What is required

We need to determine the total amount of electrical energy dissipated by the light bulb. We know the relationship between the power and energy and we are given the time. Time is not given in the correct units so we first need to convert to S.I. units:

$$\begin{aligned} 8 \text{ hr} &= 8 \times 3600 \text{ s} \\ &= 28\ 800 \text{ s} \end{aligned}$$

Step 2: Calculate the energy

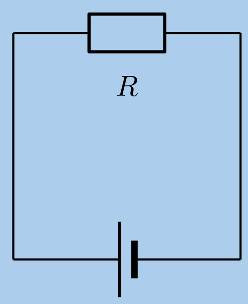
We know that:

$$\begin{aligned} E &= Pt \\ &= (30)(28\ 800) \\ &= 864\ 000 \text{ J} \end{aligned}$$

Worked example 13: Electrical energy

QUESTION

Study the circuit diagram below:



The resistance of the resistor is 27 Ω and the current going through the resistor is 3,3 A. What is the power for the resistor and how much energy is dissipated in 35 s?

SOLUTION

Step 1: Determine how to approach the problem

We are given the resistance of the resistor and the current passing through it and are asked to calculate the power. We have verified that:

$$P = I^2R$$

and we know that

$$E = Pt$$

Step 2: Solve the problem

We can simply substitute the known values for R and I to solve for P.

$$\begin{aligned} P &= I^2R \\ &= (3{,}3)^2 \times 27 \\ &= 294{,}03 \text{ W} \end{aligned}$$

Now that we have determined the power we can calculate the energy:

$$\begin{aligned} E &= Pt \\ &= (294{,}03)(35) \\ &= 10\ 291{,}05 \text{ J} \end{aligned}$$

Step 3: Write the final answer

The power for the resistor is 294,03 W and 10 291,05 J are dissipated.

Electricity is sold in units which are one kilowatt hour (kWh). A kilowatt hour is simply the use of 1 kW for 1 hr. Using this you can work out exactly how much electricity different appliances will use and how much this will cost you.

Worked example 14: Cost of electricity

QUESTION

How much does it cost to run a 900 W microwave oven for 2,5 minutes if the cost of electricity is 61,6 c per kWh?

SOLUTION

Step 1: What is required

We are given the details for a device that uses electrical energy and the price of electricity. Given a certain amount of time for use we need to determine how much energy was used and what the cost of that would be.

The various quantities provided are in different units. We need to use consistent units to get an answer that makes sense.

The microwave is given in W but we can convert to kW:900 W = 0,9 kW.

Time is given in minutes but when working with household electricity it is normal to work in hours. 2,5 minutes $= \frac{2,5}{60} = 4,17 \times 10^{-2}$ h.

Step 2: Calculate usage

The electrical power is:

$$\begin{aligned} E &= Pt \\ &= (0,9)(4,17 \times 10^{-2}) \\ &= 3,75 \times 10^{-2} \text{ kWh} \end{aligned}$$

Step 3: Calculate cost (C) of electricity

The cost for the electrical power is:

$$\begin{aligned} C &= E \times \text{price} \\ &= (3,75 \times 10^{-2})(61,6) \\ &= 3,75 \times 10^{-2} \text{ kWh} \\ &= 2,31 \text{ c} \end{aligned}$$

Activity: Using electricity

The following table gives the cost of electricity for users who consume less than 450 kWh on average per month.

Units (kWh)	Cost per unit (c)
0–150	61,60
150–350	81,04
350–600	107,43
$>$ 600	118,06

You are given the following appliances with their power ratings.

Appliance	Power rating
Stove	3600 W
Microwave	1200 W
Washing machine	2200 W
Kettle	2200 W
Fridge	230 W
Toaster	750 W
Energy saver globe	40 W
Light bulb	120 W
Vacuum cleaner	1600 W

You have R 150,00 to spend on electricity each month.

1. Which usage class do you fall into?
2. Complete the following table.

Appliance	Cost to run for 1 hour
Stove	
Microwave	
Washing machine	
Kettle	
Fridge	
Toaster	
Energy saver globe	
Vacuum cleaner	

3. For how long can you use each appliance to ensure that you spend less than R 150,00 per month? Assume you are using a maximum of 20 energy saver globes around your home.

11.4 Chapter summary ESBQH

See presentation: 243S at www.everythingscience.co.za

- Ohm's Law states that the amount of current through a conductor, at constant temperature, is proportional to the voltage across the resistor. Mathematically we write $I = \frac{V}{R}$
- Conductors that obey Ohm's Law are called ohmic conductors; those that do not are called non-ohmic conductors.
- We use Ohm's Law to calculate the resistance of a resistor. $R = \frac{V}{I}$
- The equivalent resistance of resistors in series (R_s) can be calculated as follows: $R_s = R_1 + R_2 + R_3 + \ldots + R_n$
- The equivalent resistance of resistors in parallel (R_p) can be calculated as follows: $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \ldots + \frac{1}{R_n}$
- Electrical power is the rate at which electrical energy is converted in an electric circuit.
- The electrical power dissipated in a circuit element or device is $P = VI$ and can also be written as $P = I^2R$ or $P = \frac{V^2}{R}$ and is measured in joules (J).
- The electrical energy dissipated is $E = Pt$ and is measured in joules (J).
- One kilowatt hour refers to the use of one kilowatt of power for one hour.

Physical Quantities		
Quantity	**Unit name**	**Unit symbol**
Current (I)	ampere	A
Electrical energy (E)	joule	J
Power (P)	watt	W
Resistance (R)	ohm	Ω
Voltage (V)	volt	V

Exercise 11 – 7:

1. Give one word or term for each of the following definitions:
   a) The amount of energy per unit charge needed to move that charge between two points in a circuit.
   b) The rate at which electrical energy is converted in an electric circuit.
   c) A law that states that the amount of current through a conductor, at constant temperature, is proportional to the voltage across the resistor.
2. A 10 Ω has a voltage of 5 V across it. What is the current through the resistor?
   a) 50 A
   b) 5 A

c) 0,5 A

d) 7 A

3. Three resistors are connected in series. The resistances of the three resistors are: 10 Ω, 4 Ω and 3 Ω. What is the equivalent series resistance?

   a) 1,5 Ω

   b) 17 Ω

   c) 0,68 Ω

   d) 8 Ω

4. Three resistors are connected in parallel. The resistances of the three resistors are: 5 Ω, 4 Ω and 2 Ω. What is the equivalent parallel resistance?

   a) 1,05 Ω

   b) 11 Ω

   c) 0,95 Ω

   d) 3 Ω

5. A circuit consists of a 6 Ω resistor. The voltage across the resistor is 12 V. How much power is dissipated in the circuit?

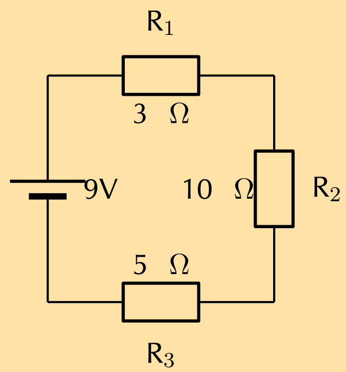
   a) 864 W

   b) 3 W

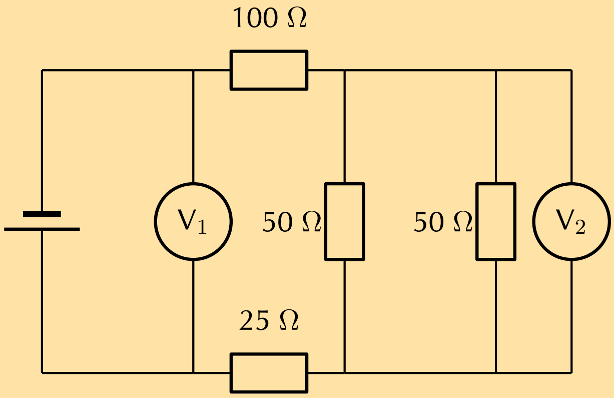
   c) 2 W

   d) 24 W

6. Calculate the current in the following circuit and then use the current to calculate the voltage drops across each resistor.



7. A battery is connected to this arrangement of resistors. The power dissipated in the 100 Ω resistor is 0,81 W. The resistances of voltmeters V_1 and V_2 are so high that they do not affect the current in the circuit.



a) Calculate the current in the 100 Ω resistor.

b) Calculate the reading on voltmeter V_2.

c) Calculate the reading on voltmeter V_1.

8. A kettle is marked 240 V; 1500 W.

   a) Calculate the resistance of the kettle when operating according to the above specifications.

   b) If the kettle takes 3 minutes to boil some water, calculate the amount of electrical energy transferred to the kettle.

   [SC 2003/11]

9. **Electric Eels**

   Electric eels have a series of cells from head to tail. When the cells are activated by a nerve impulse, a potential difference is created from head to tail. A healthy electric eel can produce a potential difference of 600 V.

   a) What is meant by "a potential difference of 600 V"?

   b) How much energy is transferred when an electron is moved through a potential difference of 600 V?

   [IEB 2001/11 HG1]

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